

Title: Context collapse amplifies opinion polarization in social media

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Abstract: I develop a mechanistic explanation of social media opinion polarization as an unintended consequence of user-initiated content retraction decisions that may reflect the relational strains of interacting with diverse social network audiences in open, context-collapsed communication environments. Empirical analysis of 17.9M U.S. Twitter users shows that users embedded in large networks with cohesive community structures retract their posts more frequently in highly visible, politically contentious communication contexts, independent of the user's own partisan position. Computational experiments using an agent-based opinion dynamics model generate substantial increases in opinion polarization at the system level, when the communication network topology mimics the context-collapsed social media environment. In contrast, pre-existing opinion polarization does not aggravate in context-preserved network structures, as the lower retraction rate, hence higher visibility, of the agent opinions that reflect the opinions of diverse neighbors, anchors the local outlier opinions more firmly from drifting to farther extremes.

Introduction

What explains the growing opinion polarization observed in social media platforms (1–3)? Computational models of opinion dynamics explain polarization as the feedback between homophily and social influence in social networks – individuals with similar opinions are relationally drawn closer to each other (homophily) and their strengthened relationship increases the influence they mutually exert on each other’s opinion (social influence), which brings their already similar opinions even closer (homophily). Under reasonable assumptions, this relational feedback mechanism is sufficient to generate opinion polarization on simulated networks of social agents (1, 2, 4, 5). Despite the longstanding challenges in disentangling homophily from social influence using observational network data (6–8), these simple models have served as a flexible tool for exploring the coevolution of network structure and opinion polarization.

A key hidden assumption of these models is that individuals can always observe the true opinions of their network neighbors. In practice, however, people frequently withhold, soften, or retract expressed opinions to avoid relational conflict, peer criticism, and social exclusion (9–12). In political contexts, individuals who perceive their views as misaligned with those of their social ties – experiencing political cross-pressure – tend to withdraw political participation to avoid relational strain (10, 13). In social media, unless experimentally induced to engage with opposing opinions (14), users tend to quietly ignore rather than publicly contest opposing views (15–19), post anonymously (20, 21), selectively disclose socially desirable opinions (22), or delete posts after publication (23–25). As a result, the opinions of outspoken users can have a disproportionately large influence on the direction of public opinion that might not materialize if the true underlying distribution of opinions were visible (26, 27).

Drawing from this rich body of work showing self-imposed sanctions in social interactions across off- and online political spaces, this paper builds an opinion dynamics model of social media opinion polarization that incorporates the systematic biases in observable opinions, arising when well-connected users disproportionately retract or self-censor their opinions to manage the relational strains in open, context-collapsed communication settings. Context collapse refers to the way social-media platforms merge otherwise distinct social audiences into one, so that an utterance is simultaneously exposed to groups a user would ordinarily address separately (28, 29). In these context-collapsed environments, outsiders (to the conversation) are likely to interpret and consume those publicly visible utterances at surface level without the situational norms, values, and interpretive frames specific to and taken for granted by those in the conversational context. Stripped of context, the opinions in these utterances can invoke negative reactions due to misunderstandings, normative objections, or competing group interests (30, 31). The user who is caught up in these micro-social frictions is subject to close scrutiny on questions of group identity and allegiances and is at higher risk of criticism, harassment, and, in extreme cases, physical threat and violence (12, 14, 32, 33).

Arguably, a user who engages with diverse audiences (i.e., large number of social groups that are clearly differentiated by few or no overlapping members) in open communication platforms is at higher risk of observing inter-audience friction and conflict when engaging with different groups. Suppose a Twitter user with followers from N nominally distinct groups (e.g., church friends, college friends, colleagues) posts a public tweet that targets one audience group. Let p be the probability that a user’s tweet targeting group i conflicts with group j ($\forall i, j \in N, i \neq j$).

j). Then, assuming p is independent across tweets and identical across group pairs, the compound risk, P , that the tweet will trigger conflict with any pair of audience groups can be described by $P = 1 - (1 - p)^{(1-\alpha)d}$ where $d = \binom{N}{2}$ possible pairs of groups and α ($0 \leq \alpha \leq 1$) is the average Jaccard similarity of the d pairs in terms of member overlap (high α corresponds to low audience diversity). α encapsulates the extent to which two groups share social similarities, since member overlap generally correlates with social proximity. As illustrated in Fig 1A, the probability of intergroup conflict triggered by a user's tweet monotonically increases with the number of the user's audience groups, weighted by intergroup dissimilarity ($1 - \alpha$). Further assuming that post retraction is a response to ameliorate the relational strain from intergroup conflict, a user's post retraction frequency should scale with the user's audience size and group differentiation.

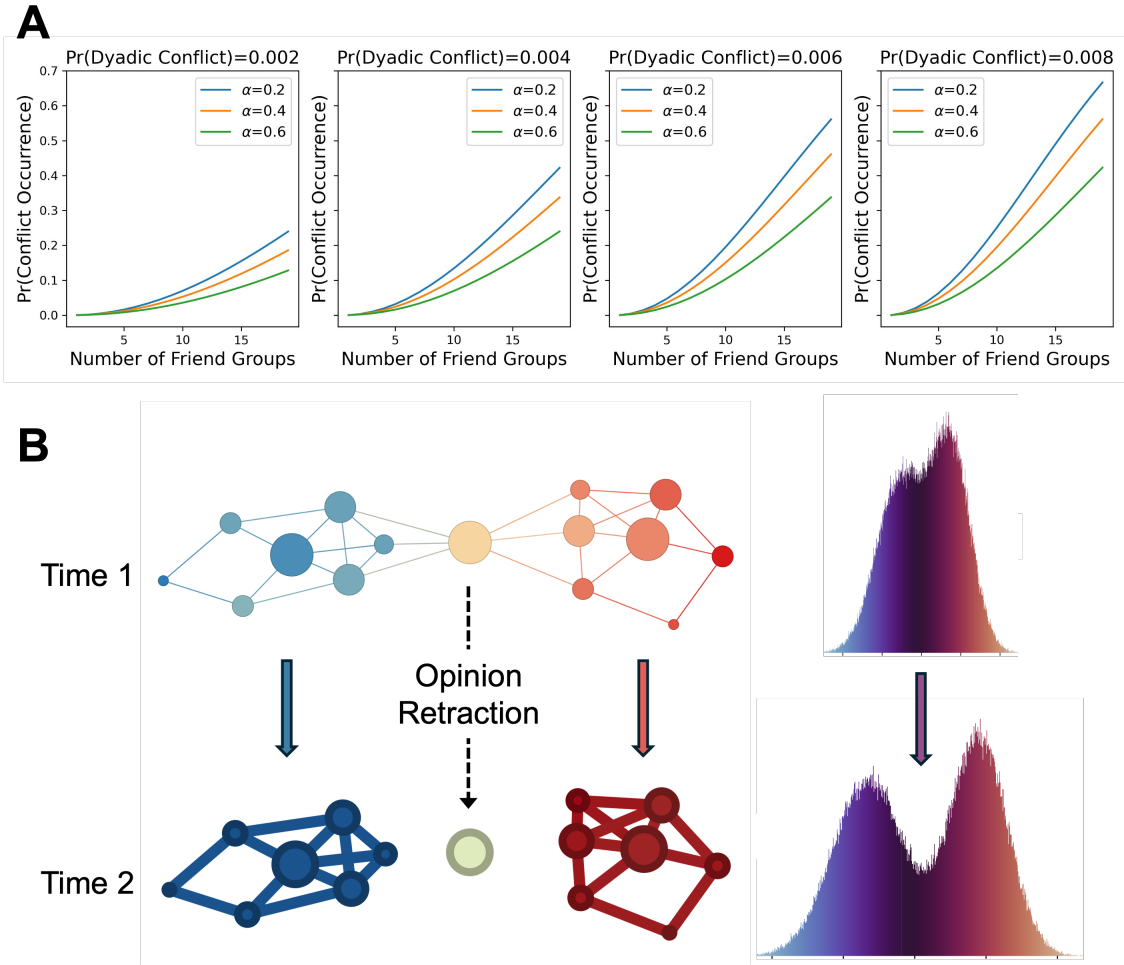


Fig. 1. Network diversity, content retraction, and opinion polarization **A.** The probability of a social media user observing conflict among their audience groups in a context-collapsed online environment increases with the number of audience groups and the extent to which the groups do not overlap in membership (i.e., low α). **B.** Network brokers (yellow) locally bridge diverse clusters and anchor extreme opinions (Time 1); retraction of their opinions to avoid relational strain can deepen polarization (Time 2).

Ironically, the higher probability of opinion retractions by users who interact with diverse groups can potentially lead to greater intergroup conflict at the macro level. This is because these are the users who are also well-positioned to bridge otherwise disconnected groups – translating, reconciling, or integrating opposing viewpoints between groups. As well-connected opinion leaders, they can “divide-and-conquer” (34, 35) – carefully reframing one group’s view (e.g., gay rights) with the moral values that resonate in the context of the opposing group (e.g., the rights of a loyal patriot who happens to be gay) (36). However, bridging social groups often requires interacting with each in isolated context. In the open social media environment, these users may quickly retract their balanced, reconciliatory opinions upon discovering the loss of context and nuance. Collectively, the under-representation of balanced views, or the over-representation of the remaining extreme views can drift people’s opinions to farther extremes, aggravating opinion polarization at the macro-level (Fig. 1B). While offline spiral of silence, driven by those who believe they hold the minority opinion, tends to result in public consensus (11, 37), its online counterpart, driven by potential opinion leaders, could end in more extreme polarization.

Results

Measuring the diversity of a Twitter user’s communication network requires a broad record of user interactions. To this end, I constructed a topic-agnostic communication network of 26 million U.S. users from a large, user-centric 2014 crawl of timelines with the most recent 3200 tweets per user (38). Despite its age, this crawl is uniquely suited for the present analysis because it captures user communications across diverse audiences at population scale. Unlike most topic- or event-centric research datasets that record user interactions within narrowly defined study contexts, it provides the rare opportunity to observe users’ broad timeline activity that spans multiple social and conversational contexts (SI section 1.1). I then joined this network with user-initiated tweet-deletion counts from Twitter’s Decahose over 2011–2018, yielding counts for 17.9M filtered accounts (SI section 1.2).

I quantified the diversity of a user’s communication network structure with three measures: (a) unweighted node degree — the size of a user’s mutually engaged social media audience; (b) degree-adjusted Shannon entropy of tie weights (39, 40) — the evenness with which the user’s engagement spreads across that audience; and (c) the weighted clustering coefficient (41) — the share of the user’s interaction volume occurring within shared social contexts. A neighbor pair that also interacts with each other indicates a context common to the user and both neighbors, within which the user’s posts are more likely to carry mutually understood meaning. Hence, the weighted clustering coefficient measures heavily contextualized communications in distinct interconnected groups, the content of which is, arguably, most exposed to misreading when the platform delivers it beyond the context it was written for. Empirically, all three measures showed medium to high correlation with Burt’s network constraint index (42), the broadly used composite measure of an individual’s structural brokerage potential (Table S1). See SI section 1.5 for construct validity.

Opinion Retraction and Network Diversity

While a social media user's active engagement with diverse network neighbors can create brokerage opportunities for information arbitrage and social influence, it also entails the risk of being mired in the relational conflicts that would limit a user's self-presentation and opinion expression. The model-estimated marginal effects of network diversity on tweet deletion frequency for the 17.9M U.S. Twitter users exhibit mixed patterns that would suggest the countervailing benefits of structural brokerage and strains of navigating the competing expectations held by diverse network neighbors (Model 2, Table S2). First, the users whose interactions were spread more evenly across network neighbors (i.e., Shannon entropy) tended to delete more tweets, although the effect was moderate (Fig. 2A). Second, the deletion frequencies were generally lower for users who communicated with larger numbers of neighbors (i.e., node degree), reflecting the structural advantages that come from size (Fig. 2B). However, the degree – deletion association flipped positive for those who were embedded in the most clustered networks (degree-clustering interaction $B=0.11$), as illustrated by the users in the top quartile of weighted clustering in Fig. 2B. These users, whose large audiences are organized into dense shared contexts – in the context-collapsed environment of Twitter – deleted most heavily. Finally, users whose network interactions were more concentrated in clusters of neighbors (i.e., high weighted clustering — heavily contextualized interaction) tended to delete more tweets and, mirroring Q4 in Fig. 2B, the increase in deletions was greater for users in the top degree quartile (Fig. 2C).

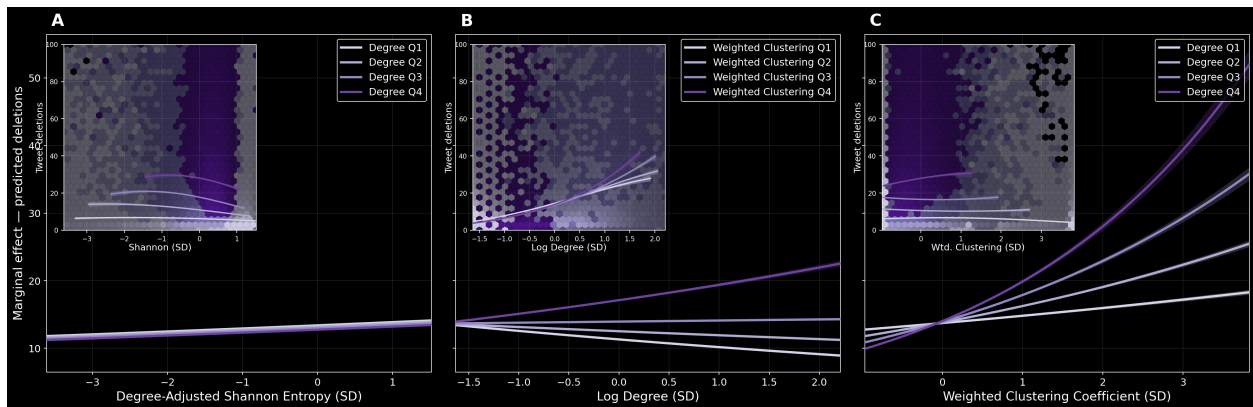


Fig. 2. Predicted tweet deletions by three network diversity measures (full sample) – (A) degree-adjusted Shannon entropy of tie strengths, (B) logged node degree, and (C) weighted clustering – using standardized regression coefficients estimated from the full sample of 17.9M users (Model 2, Table S2). Each panel shows the model-predicted marginal effect (95% CI band), with the stratifying variables (degree/clustering) held at the median value of each stratum and the other covariates held at their respective sample medians. **Inset. Each curve in an inset is a within-stratum second-order polynomial fit (95% CI from 200 bootstrap resamples) of the model-predicted deletion values using each user's observed covariates. The curves are overlaid on a hexagonal-bin plot that shows the bivariate distribution of the plotted variables. The color of a hexagonal bin corresponds to the stratum with the highest within-stratum proportion in that bin and its brightness monotonically increases with this within-stratum proportion.**

These mixed and relatively weak associations in the full sample became directionally more salient for a subsample of 184K politically active U.S. users whose partisan positions were inferred from the political figures they followed on Twitter (43, 44) (see SI section 1.3). For these users who were arguably more exposed to political contention on Twitter, the effect sizes (standardized coefficients) of the three network diversity measures increased – Shannon entropy and node degree increased from 0.04 to 0.068 and 0.00 to 0.1 respectively, while weighted clustering decreased from 0.2 to 0.156 (Model 1, Table S3). Although the political moderates maintained marginally more diverse networks than the partisans, as measured by Burt’s network constraint (Fig. S1A), the partisans actually retracted substantially larger volumes of tweets than the moderates (Fig. S1B). However, the associations between post retraction and network diversity were robust across the extremity spectrum — centrist, moderate, and extreme users alike (Fig. 3D-3F; Table S3 Model 3 augmented with $|\text{polarity}| \times \text{network-measure}$ interactions). These associations also held positive net of the partisan misalignments among a user’s neighbors – a potential source of relational strain and conflict – as measured by the median polarity score distance between neighbors (see Model 3, Table S3 for detailed estimates). Taken together, these results highlight the network-structural diversity as a potential source of relational strain that cannot be accounted for by individual political orientation or by political cross-pressure (10, 13) within one’s network neighborhood.

Granted, post retraction may not necessarily reflect the relational consequences of posting about hot-button issues or invoking identity claims that trigger conflict before an indistinct audience (33). The data quantify this concern. For a subsample of 25,225 polarity-score-matched users, 12.4% of posted tweets were classified as political (SI section 3). Deletions are observable only through the Decahose’s random 10% stream of deletion events, which identifies 0.018% of these users’ 36 million tweets as deleted — an observed rate implying roughly ten times as many actual deletions. Of the observed deletions, 7.4% were political at the user level, and at the tweet level political tweets were deleted slightly less often than non-political ones (SI section 3). The sparsity of political tweet postings and deletions makes it difficult to isolate tweet-level political correlates that may account for retraction decisions within the partisan structure of a user’s network (33).

Nevertheless, user-level retraction patterns can still prove useful in narrowing down alternative explanations of post retraction. I triangulated with two complementary tests to rule out a prominent non-political source of post retraction variation: an eliminative test that turns one plausible non-political account of retraction into a falsifiable prediction, and a confirmatory test that traces the retraction–diversity association to the partisan composition of the audience. First, the eliminative approach rests on the assumption that if retractions were driven by general curatorial considerations (e.g., accuracy, normative adequacy, and content-audience fit (19, 45, 46)), rather than by relational concerns centered specifically on political cross-pressure and conflict aversion (28), a user would retract more often if the general content characteristics of their tweets are (a) too “quirky” or personal relative to the broader population of users (i.e., “content atypicality”) and/or (b) too niche to appeal to a broad range of non-overlapping interests of the network neighbors (i.e., neighbor-neighbor content diversity). If the deletions do not increase with the user’s content atypicality and/or the neighbors’ content diversity, their retraction decisions in the aggregate are difficult to attribute to general curatorial considerations. This would not confirm that deletions are motivated by conflict aversion or political cross-pressure directly, but it would rule out the content–audience-fit explanations.

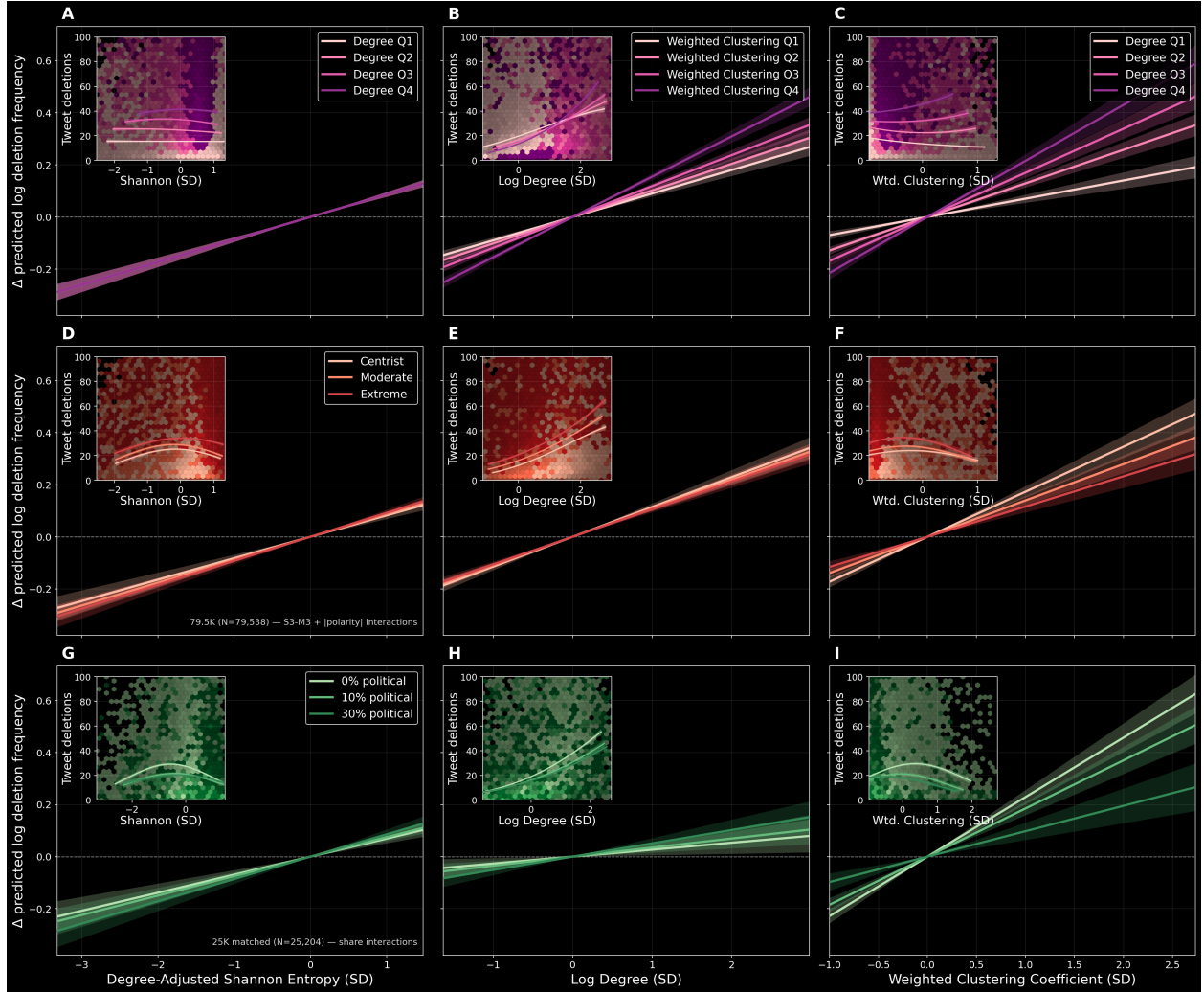


Fig. 3. Predicted tweet deletions by three network diversity measures and political partisanship – (A, D, G) degree-adjusted Shannon entropy of tie strengths, (B, E, H) logged node degree, and (C, F, I) weighted clustering – using standardized regression coefficients estimated from a subsample of 80K (25K for G-I) politically engaged Twitter users (A–C: Model 4 of Table S3, $N = 79,538$; D–F: Model 3 of Table S3 augmented with $|polarity| \times$ network-measure interactions; G–I: Model 2 of Table S3 augmented with political-tweet-share $\times$ network-measure interactions, estimated on the 25K classified subsample, $N = 25,204$). Each panel shows the model-predicted marginal effect (95% CI band), with the stratifying variables (degree/clustering for A–C; extremity ($|polarity|$) for D–F; proportion of political tweets for G–I) held at the median value of each stratum and the other covariates held at their respective sample medians. The centrist, moderate, and extreme categories correspond to the bottom, middle, and top thirds of the absolute polarity score respectively; the political-tweet-share strata correspond to shares of 0%, 10%, and 30%. **Inset.** Each curve in an inset is a within-stratum second-order polynomial fit (95% CI from 200 bootstrap resamples) of the model-predicted deletion values using each user's observed covariates. The curves are overlaid on a hexagonal-bin plot that shows the bivariate distribution of the plotted variables. The color of a hexagonal bin corresponds to the stratum with the highest within-stratum proportion in that bin and its brightness monotonically increases with this within-stratum proportion.

I measured the general content characteristics of a tweet by embedding each tweet into a 384-dimensional vector using Sentence-BERT (47) for two samples – a random subsample of

14K users from the full 17.9M user data and the 25K users in the polarity score-matched sample drawn for the above-mentioned political tweet classification (SI section 3). The tweet embeddings were then aggregated to represent a user’s content characteristics. Subsequently, a user’s general content atypicality was measured as the cosine distance between the focal user’s aggregate vector and the centroid of all user-level embeddings. Similarly, the content diversity of the neighbors was measured as the median cosine distance among the neighbors’ mean vectors. See SI section 4 for procedures and a qualitative analysis of the content characteristics.

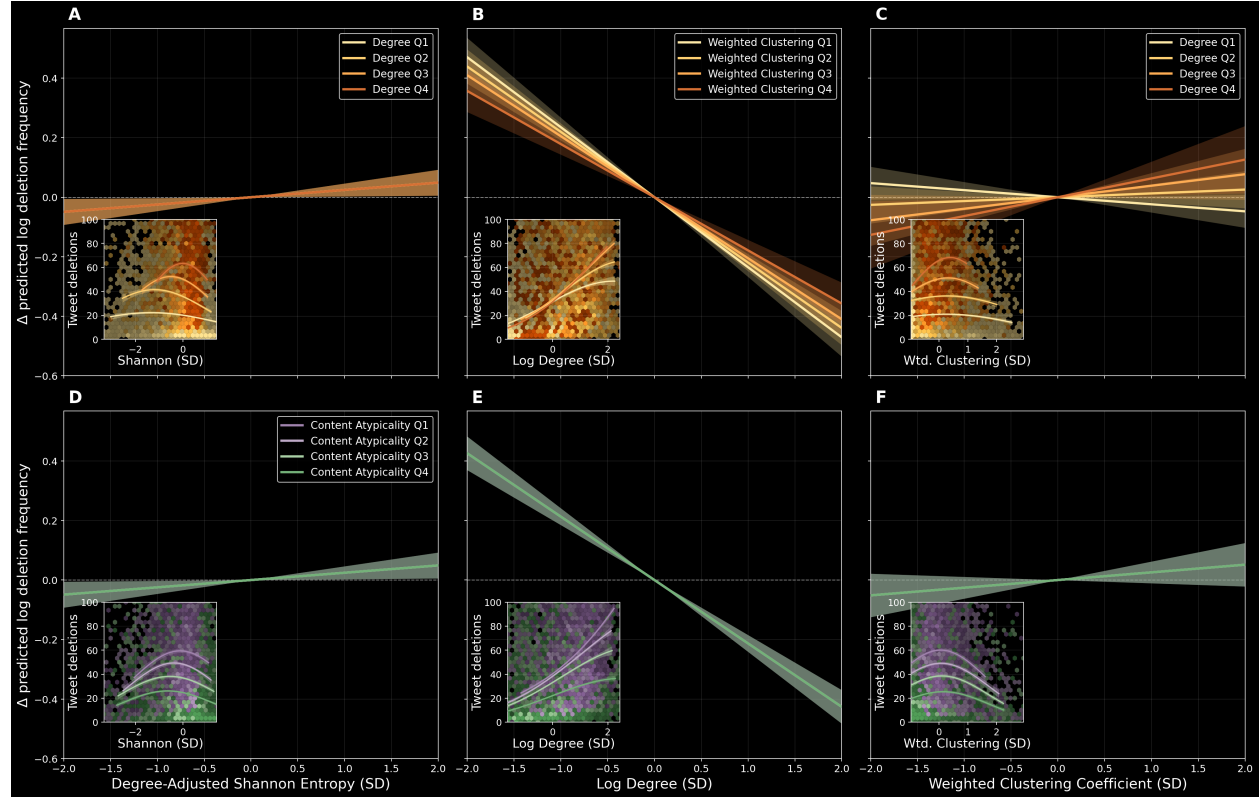


Fig. 4. Predicted tweet deletions by three network diversity measures and tweet characteristics – (A, D) degree-adjusted Shannon entropy of tie strengths, **(B, E)** logged node degree, and **(C, F)** weighted clustering – using standardized regression coefficients estimated from a subsample of 14K Twitter users whose content characteristics were quantified by embedding their tweets into a 384-dimension vector using Sentence-BERT (Model 4, Table S4). Each panel shows the model-predicted marginal effect (95% CI band), with the stratifying variables (degree/clustering/content atypicality) held at the median value of each stratum and the other covariates held at their respective sample medians. Content atypicality is measured by the cosine distance between the mean tweet vector of a user and the centroid of the full embedding space. **Inset.** Each curve in an inset is a within-stratum second-order polynomial fit (95% CI from 200 bootstrap resamples) of the model-predicted deletion values using each user's observed covariates. The curves are overlaid on a hexagonal-bin plot that shows the bivariate distribution of the plotted variables. The color of a hexagonal bin corresponds to the stratum with the highest within-stratum proportion in that bin and its brightness monotonically increases with this within-stratum proportion.

Findings indicate that tweet deletions exhibited by these general users were likely driven neither by political conflict aversion nor by interest/topic-based content curation (Fig. 4). First,

the deletion patterns in light of the random users' network structure (Model 4, Table S4) sharply deviated from the politically engaged users (Model 4, Table S3) in ways that do not neatly fit a political conflict aversion account of post retraction – Shannon entropy remained positive, but the effect size was less than a third of that in the polarity-score-matched sample ($B=0.025$ vs. $B=0.087$, visualized in Fig. 4A); the degree term unambiguously flipped negative (Fig. 4B) and the clustering term collapsed once the content characteristics were controlled ($B=0.007$ vs. $B=0.073$). These partial associations tended to diminish after adjusting for the user and neighbor content characteristics (Fig. 4D-4F). Nevertheless, these deletion patterns were not consistent with the content-curatorial account either – the users' content atypicality (Fig. S1E) and the content diversity of their neighbors (Fig. S1F) both showed negative associations with tweet deletions. The same patterns were confirmed in the polarity-matched sample as well (Model 2, Table S5) – their post retractions declined with both content atypicality ($B = -0.059$) and the median neighbor-neighbor embedding distance ($B = -0.262$), suggesting that retraction may not reflect general curatorial concerns regardless of political engagement.

On the other hand, the confirmatory test asks whether the retraction–diversity association carries the specific signature of political cross-pressure: if retractions reflect the relational strain of catering to a politically mixed audience, then (a) retractions would generally increase with growing partisan dispersion of a user's network neighbors, but more strongly for politically centrist/moderate users caught between adversarial camps. Furthermore, (b) these predicted patterns should survive controls for the general (non-political) content characteristics of the user and their audience.

In the polarity-scored sample (Table S3, Model 4), the median neighbor–neighbor polarity distance was positively associated with retractions ($B=0.102$) and the association was indeed stronger for the centrists and moderates (Fig. S2A). Furthermore, the same pattern was observed at nearly identical magnitudes after controlling for content atypicality and the median NN-embedding distance: for a subset of 10,867 users whose NN-polarity distances were observable, the association for the centrists was positive, but turned negative for the extremists as shown in Fig. S2 ($B = 0.099$, $t = 4.9$ vs. $B = -0.063$, $t = -4.5$; Table S5, Model 4). In short, the centrists and moderates deleted fewer tweets than the liberals and conservatives farther from the middle in absolute numbers, but showed steeper retraction gradient with respect to the neighbors partisan misalignment, possibly because they are the most active self-censors who refrain from posting politically sensitive content in the first place, but also delete more aggressively when they do.

Although political self-censorship is unobservable, the deletion-network diversity gradient differences between the politically vociferous vs. the reserved users are consistent with possible self-censorship even in this politically engaged group. Predicted retractions generally increased with the increasing network diversity across different levels of the share of a user's political tweets in their tweet corpus, but the association was stronger for users with smaller shares (Table S6). The pattern was most visible in the deletion-clustering gradient (Fig. 3I) where the most politically reserved users (i.e., 0% political tweet stratum) were predicted to be the most aggressive deleters.

Opinion Polarization

Individuals connected to diverse network communities tend to occupy structural brokerage positions where those diverse connections could be used strategically to reconcile or integrate competing opinions across communities (35, 48, 49). However, as the content retraction patterns suggest, context-collapsed communication environments can undermine these integrative functions of network brokers. When brokers retract their posts to avoid relational strain arising from collisions of interests, beliefs, and values (50, 51), their network neighbors get less exposure to—and less influenced by—these integrative views.

I tested the generative implications of retractions for opinion polarization using a simple opinion dynamics model (1, 4, 5, 52–54). This model simulates a social media environment in which networked agents update their opinions under their neighbors' influence and concurrently strengthen (weaken) ties to neighbors holding similar (dissimilar) opinions (i.e., homophily). In most models, agent opinions are always visible to the network neighbors at every simulation time step. The present model relaxes this assumption by allowing agents to decide at every time step whether to retain or retract (i.e., delete) their posts with the opinions updated. Retraction decisions depend on the observed opinion difference between a focal agent and its neighbors (ego–neighbor difference) and the opinion differences among neighbors (neighbor–neighbor difference), based only on the observable neighbor opinions in the previous step. As a result, the neighbors' retraction decisions indirectly affect the focal agent's opinion trajectory over time (SI section 5.2).

I conducted computational experiments using this model in a factorial design comprising 59,400 parameter combinations (see SI section 5.3). Fig. 5 summarizes the agent-level predicted post retraction probabilities by the three network diversity variables, similar to Fig. 3. The model generated patterns consistent with those observed from the politically engaged Twitter users. Among the three network variables, node degree showed a strong positive association with post retractions (Fig. 5B) and the moderate/centrist agents tended to retract fewer posts than the partisan agents (Fig. 5D–5F). The agents with diverse networks were also the model's moderates: within runs, the correlation between an agent's degree-adjusted Shannon entropy and its opinion extremity averaged -0.54 , nearly invariant across all six network types, whereas node degree and clustering were unrelated to extremity (Fig. S10). The model also deviated from the empirical results in other respects - Shannon entropy (Fig. 5A) and clustering (Fig. 5C) exhibited weak or negative correlations with retraction probability. This discrepancy stems from the model's exclusive focus on the relational constraints (i.e., cost) of network diversity with no mechanism to incentivize agents to seek diversity (see SI section 5.4 for elaboration).

I examined three run-level macro-outcomes – mean probability of retraction; true and observed polarization, which were measured as the percent-increase in the Esteban–Ray (ER) polarization index (55) at the end of a simulation run, relative to the initial state. True polarization is measured using every agent's opinion at the final simulation step, whereas observed polarization uses only the opinions that agents did not retract in the final step. Enabling retraction is de-polarizing on average – relative to otherwise identical runs with retraction disabled, retraction-enabled runs show systematically smaller polarization growth (mean true-polarization change $+77\%$ vs. $+119\%$; medians $+22\%$ vs. $+57\%$). This population-level average, however, conceals a sharp dependence on network structure where retraction becomes polarizing.

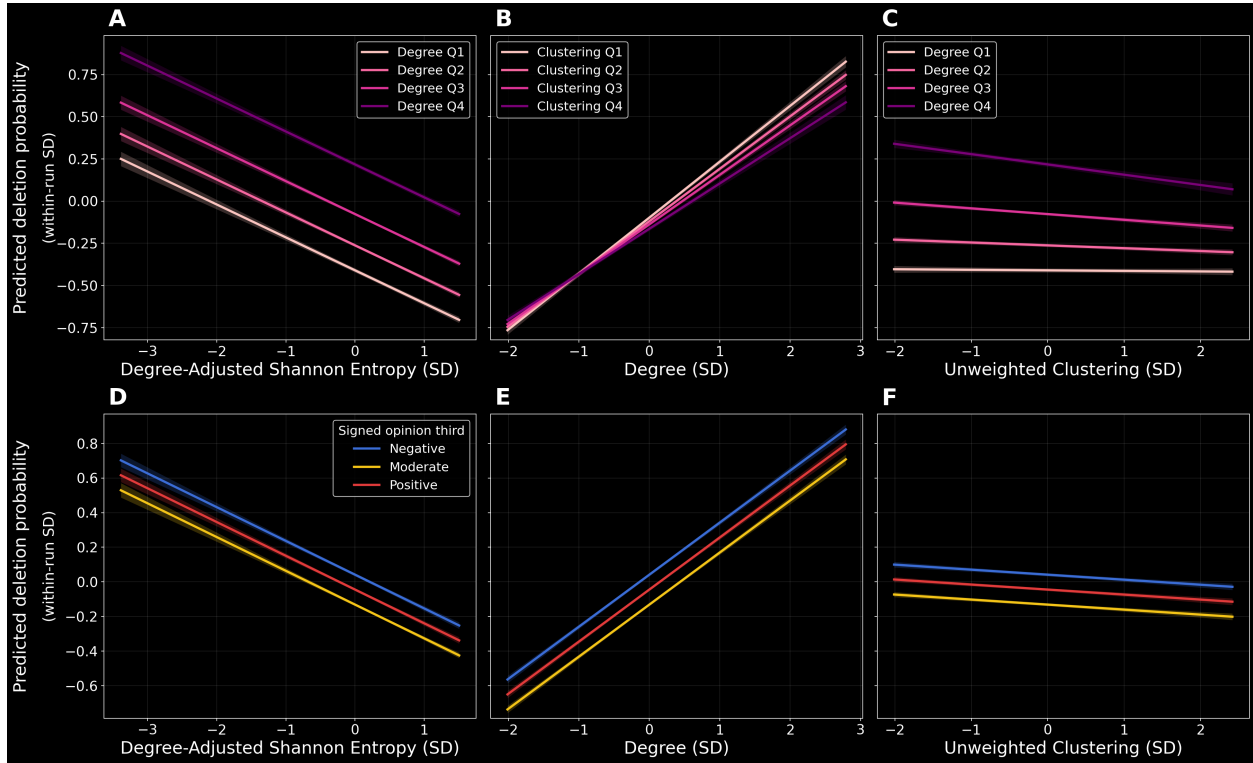


Fig. 5. Post retraction probability from the agent-based opinion dynamics model. Slopes show marginal effects of three network-diversity measures – degree-adjusted Shannon entropy, node degree, and unweighted (topological) clustering – on post retraction probability (with 95% CI bands) based on Model 2 of Table S7. Row 1 (A–C) stratifies by degree/clustering quartiles; Row 2 (D–F) stratifies by the agent's signed opinion into thirds (negative / moderate / positive), the simulation analogue of the liberal / moderate / conservative split in Fig. 3D-3F. Predictors are z-scored within each run.

The six network types (Fig. 6D-6F) represent stylized communication environments that vary in degree distribution, local clustering, and, most central to the experimental manipulation, graph density. Higher density graphs collapse distinct local communities into fewer audience groups with the completely connected network (clique) being the limiting case. A post in a clique reaches the maximal possible audience at once, which stylistically mimics a tweet reaching far beyond its author's immediate network neighbors through retweets and algorithmic recommendation; the stylization is that this reach is treated as immediate and complete rather than unfolding in waves such as retweet cascades.

Agents in the clique structure retracted with far higher probability (Fig. 6D) and both true and observed polarization rose the most (Fig. 6E, 6F); the five other networks that preserve local structure exhibit rather modest changes in polarization (e.g., Fig. S9 for clustering). The stark separation of the clique from the rest—and the limited variation among the other networks—indicates that it is simple graph density, above other structural graph features, that turns retraction into a polarizing force. Incremental increases in graph density shown in Fig. 6G-6I reiterate this point – both retraction and polarization increase monotonically as mean degree $\langle k \rangle$ increases from 24, 48, to the clique's mean of 199. As a network densifies, each agent is exposed

to a broader range of opinions with mechanically increasing conflict probability as their node degree increases (Fig. 5B).

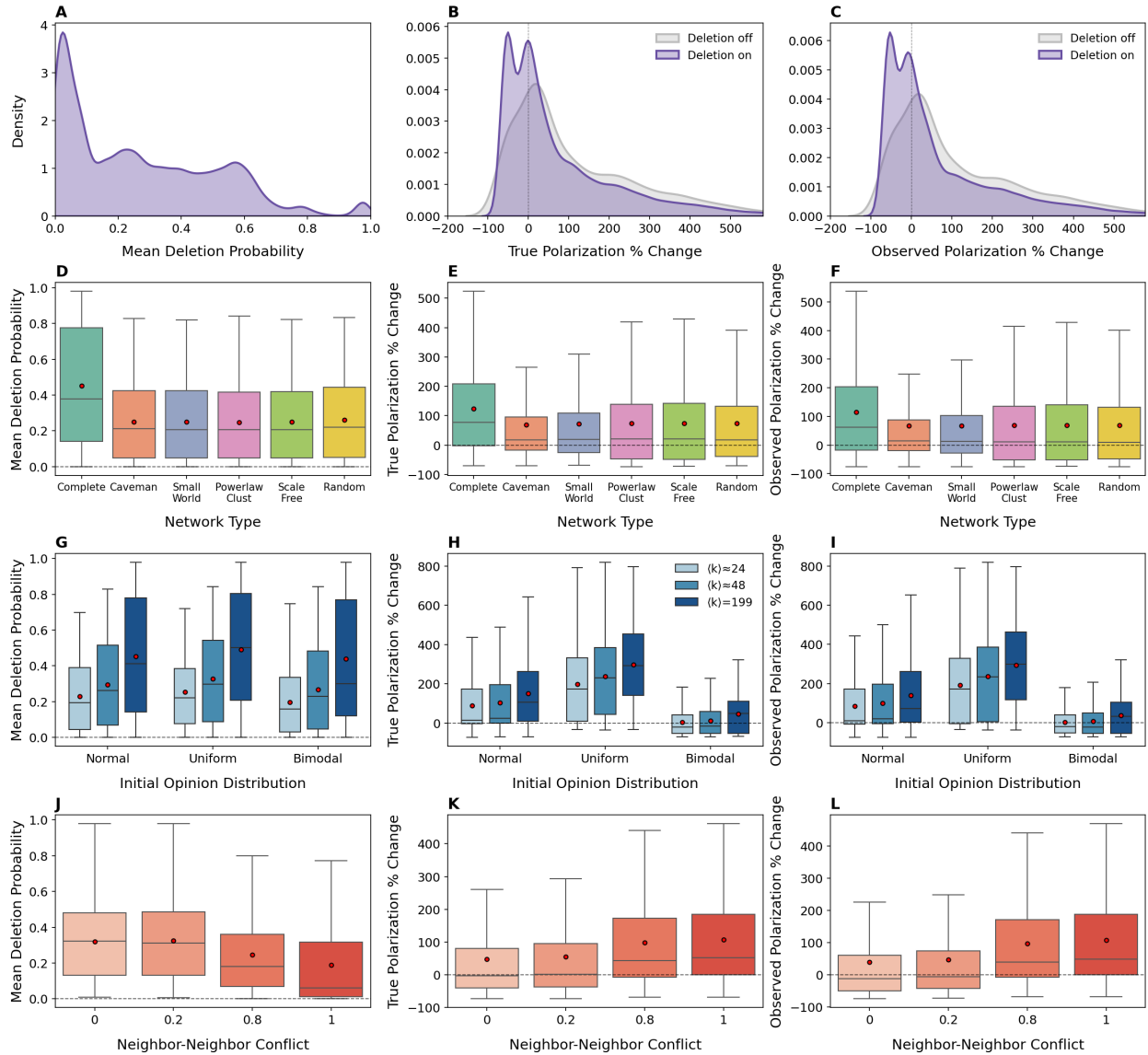


Fig 6. Distributions of three macro-outcome measures from the computational experiments – deletion probability (column 1), percentage change in true (column 2) and observed (column 3) opinion polarization from the initial to the end state. Polarization is measured with the Esteban–Ray metric. A.–C. Kernel density estimates for simulation runs where post deletion is enabled (purple) vs. disabled (gray); the deletion-probability panel (A) is shown for enabled runs only. The three macro-outcomes are stratified by network type (D.–F.), by initial opinion distribution and mean degree (G.–I.), and by the weight that agents place on interneighbor conflict relative to ego–neighbor conflict in their post retraction decisions (J.–L.).

The empirical analysis showed that the politically engaged users retracted more posts with growing partisan misalignment of their neighbors. I tested the macro implications of this

user-level behavior using the parameter β_a in the model's retraction decision function (Eq. 1 in SI section 5.2), which determines the weight that agents place on the opinion disagreements that the agent observes among its neighbors (neighbor–neighbor distance) relative to the disagreement between the agent's own opinion and the observable opinions of its neighbors (ego–neighbor distance). At $\beta_a = 0$, agents retract only in response to their own clashes with their audience and completely neglect inter-neighbor conflict; at $\beta_a = 1$, only in response to division within the audience. The last row of Fig. 6 sweeps this weight, and it separates how much retraction occurs from who retracts. Mean retraction declines as the weight shifts toward audience division (Fig. 6J): division-triggered retraction is more selective, not more frequent. True polarization nevertheless rises steadily with the weight (median true-polarization change -2% , $+1\%$, $+43\%$, and $+52\%$ at $\beta_a = 0, 0.2, 0.8$, and 1.0 ; Fig. 6K–6L). The two gradients move in opposite directions: as retraction shifts from the agent's own conflicts toward the divisions it witnesses, fewer posts are retracted, yet polarization grows. This is because the ego–neighbor channel selects locally deviant opinions for removal — a moderating selection that holds polarization down — whereas division-triggered retraction progressively withdraws that selection and shifts silencing toward the agents exposed to divided audiences. The macro-outcome thus tracks not the amount of self-censorship but its target — the same selective silencing the empirical analyses show for cross-pressured users.

The polarization results reported in Fig. 6 are robust to the following modeling assumptions. First, the initial opinion distribution the magnitude of the effect of graph density on resulting polarization. However, it does not qualitatively alter the outcome as shown in Fig. 6G–6I. Second, the results do not depend on the initial edge weight distribution (Fig. S3) and are robust to the level of stubbornness η_1 and susceptibility η_2 parameters in the opinion update function Eq. 7 (Fig. S4). Together, these experiments show how the same retraction behavior that moderates opinion in context-preserving networks can exacerbate it under context-collapse.

Discussion

Post retraction is a form of self-presentation—a corrective response to both anticipated and realized relational tensions arising from interactions with an audience. In this sense, post retractions are motivated by the same self-presentational concerns that underlie social media self-censorship (18, 19, 45, 56). Although retractions are not, by definition, acts of self-censorship, they reveal conflicting social boundaries that users may initially underestimate or fail to recognize (57), but subsequently learn to anticipate and manage through self-censorship (19).

While most users attempt to maintain a consistent self-image, the results of this study suggest that diversity in one's interaction network complicates self-presentation, increasing the likelihood of misjudgment, as reflected in network brokers' frequent post retractions. It is therefore plausible that these users also self-censor more frequently, although self-censorship itself remains difficult to measure directly using observational data.

In the explanatory framework developed here, the absence of context separation is a key condition that can lead to online polarization. The technological affordances that facilitate open information exchange in contemporary social media platforms (e.g., timelines, algorithmic recommendations, trending topics/posts, following and reposting functionality) have enabled

rapid information diffusion (58, 59) and new forms of decentralized coordination and collective action (60–62). Indeed, among the majority of Twitter users who are not heavily involved in political discourse, users with diverse networks and/or those with atypical content characteristics appear less inhibited in their self-expressions (Fig. 4).

However, the flip side of this low-friction, high-velocity communication environment is that it blurs situational contexts and obscures the shared frames of reference that need time to develop through repeated social interaction (56, 63–65). It is this uncertainty that can be particularly detrimental to social integration where social integration is most critical – politics. At the individual level, the network brokers, who are well-positioned to bridge disconnected groups and reconcile opposing opinions (35, 48, 49, 66) are arguably burdened the most by the same technological affordances that have accelerated information diffusion – they must navigate the risks of sharing opinions stripped of nuance with network neighbors who may disagree on hot-button issues (Fig. 3). In short, the very conditions that enable large-scale information exchange and social organization may also undermine the integrative social functions traditionally performed by network brokers.

Although the empirical analysis of this study does not causally identify the effects of context collapse on online opinion polarization, results from the virtual experiments (Fig. 6E and 6H) suggest that channeling information flow on the basis of adequate context separation could attenuate existing partisan divisions. In practice, interventions aimed at reducing polarization and extremism have focused primarily on the consumption side – on curating what information is prioritized for recipients – through algorithmic recommendation, downranking, and moderation strategies designed to limit toxic partisan speech within filter bubbles (67–70). However, the simulation experiments imply that a consumption-side approach may be insufficient if network brokers’ content retractions and potential self-censorship diminish the supply of balanced, integrative opinions that better reflect the underlying local opinion distribution.

This limitation calls for attention to the production side of information channeling, with an emphasis on granting users greater visibility and control over the context of their own interactions. An early effort in this direction was the *Circles* feature of the now-defunct social media platform Google+ that allowed users to custom-assemble their audiences rather than rely on rudimentary audience categories based on social distance (e.g., private, friends-only, public) or functional roles (e.g., friend, family, work) (71, 72). Although subsequent implementations of production-side approaches—and related research on supporting social media audience selection—have been limited (73–75), the results of this paper point to the need for privacy-enhancing audience control tools capable of flexibly separating audience contexts.

Materials and Methods

Data and empirical analysis. The empirical analyses use a topic-agnostic communication network of 26M U.S. Twitter users, constructed from a 2014 user-centric crawl that retrieved up to 3,200 recent posts for each of 158M users worldwide, with bidirected ties defined by mutual @mention and @reply tweets (SI section 1.1). User-initiated tweet deletion data from Twitter’s Decahose (a random 10% real-time stream) over 2011–2018 were joined with the mention network, resulting in 17.9M users after excluding the top 5.5% of accounts by deletion volume

(≥ 351 deletions) and by total activity ($> 20,000$ posts) to remove unusual bulk-deletion accounts and non-individual accounts (SI section 1.2). User-level network diversity was quantified with three complementary measures that moderately correlated with Burt's network constraint index – logged node degree ($r = -0.77$), the degree-adjusted Shannon entropy of tie weights ($r = -0.5$), and the weighted clustering coefficient ($r = 0.3$) (SI sections 1.4–1.5). The log of tweet-deletion frequency was modeled with these diversity measures, their interactions, and behavioral controls using OLS with standardized coefficients (Tables S2–S6). The results in Figs. 2–4 summarize the results with model-predicted marginal effects of each diversity measure, along with an inset that shows the aggregate (and smoothed) predicted values based on the users' observed covariate values, stratified by other network diversity measures. The analyses were repeated on nested subsamples that add (a) a continuous measure of user partisanship position (liberal – centrist – conservative) inferred from the accounts a given user follows (SI section 1.3), (b) the share of machine-labeled political tweets in a user's timeline data, based on a DeBERTa-v3 classifier fine-tuned on 4,000 hand-annotated political tweet labels (SI section 3), and (c) vector representation of a user's content characteristics, obtained by using Sentence-BERT (all-MiniLM-L6-v2) to embed each of their tweets and aggregating them within user (SI section 4).

Agent-based model. To assess whether individual-level retraction can generate macro-level opinion polarization, I extended a standard class of network opinion-dynamics models in which agents co-evolve opinions and tie weights through social influence and homophily. The opinion of a focal agent (ego) shifts closer toward the opinions of their strong tie neighbors (attraction) and pushed farther away from the opinions of their weak tie neighbors (repulsion). Through a similar process, these opinion changes subsequently affect the agent's tie weights that either grows stronger or weaker based on the similarity between the agent and the neighbors in their respective updated opinions. In the current extension to this attraction-repulsion model, each agent probabilistically retracts (deletes) its post at each time step as a function of both ego–neighbor and neighbor–neighbor opinion distances, so that the updated opinion is observable to its neighbors only if the agent does not retract (SI sections 5.1–5.2). I conducted an in-silico experiment using this model over 59,400 combinations of model parameters including, among others, communication-network structure (degree distribution, average clustering, and density), initial opinion distribution, and a parameter that determines the relative weight given to neighbor–neighbor over ego–neighbor opinion distances in the retraction probability calculation (SI section 5.3). The levels of opinion polarization were measured at the beginning and at the equilibrium state of each run using the Esteban–Ray polarization metric. Fig. 6 quantifies the aggravation/attenuation in polarization in terms of the percentage change in the Esteban–Ray metric for the entire population (true polarization) and for the subset of observed opinions at the last simulation step (observed polarization) (SI section 5.4).

This study was exempted from Carnegie Mellon University IRB review.

Data Availability. The agent-based model source code, its output data, and the empirical analysis code are provided with this submission. Upon acceptance for publication, the deidentified network data and the tweet deletion records, along with the relevant analysis scripts, will be deposited to public repositories, including the Harvard Dataverse and GitHub.

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Supplementary Information

1. Data and Measurement

1.1. Mention Network Data

The bidirected Twitter mention network largest component of 26M U.S. users used in this study was constructed from a large-scale snowball sample of user timelines (158M users worldwide) collected from November 2013 to September 2014 for a separate study (38). The data collection prioritized volume and coverage, such that the crawler ran until no new user could be found (coverage) and for each user, up to 3200 most recent tweets and retweets, which the Twitter API allowed, collected (volume). Therefore, the mention network constructed from this dataset is based on all pre-2015 mention tweets that could be found within the API's constraints. As a result, the ties in this network represent a broader set of relationships compared to the narrowly constructed political communication ties around specific issues (e.g., Obamacare), which the vast majority of studies use. The wide variety of communication ties is essential for accurately measuring network brokerage and the potential intensity of context collapse.

1.2. Tweet Deletion Data

Tweet deletion records between 2011 and early 2018 were collected from the Twitter Decahose (10% real-time tweet stream). A deletion record is a short JSON-formatted string broadcasted to Decahose subscribers in real time, containing the tweet ID, user ID, and time of deletion. As shown in fig. S5, the number of these deletion records relative to the monthly total volume of tweets streamed through the Decahose steadily increased during this period. From these real-time deletion records, I aggregate the total number of tweet deletions by each user between 2011 and 2018 and use this count as a measure of content retraction (fig. S6). Among the 26M U.S. Twitter users, 19.7M deleted a total of 2.09B tweets in the span of 2011 to 2018. Research has shown that a small proportion of accounts themselves delete a large number of tweets and that these deletions can be abused for spreading disinformation, flooding promotional content, and inflating account popularity metrics while averting detection (76). Furthermore, Twitter also suspends malicious accounts, which leads to the bulk deletion of their tweets (23). I therefore filter out the top 5.5% user accounts in total tweet deletions (351 or more deletions). I further filter out the top 5.5% of user accounts with the highest "statuses_count" (i.e., more than 20 thousand tweets and retweets) to exclude potential non-individual accounts (bots, organizations) with unusually high Twitter activity. After filtering, the dataset consists of 17.9M accounts with a total of 618.8M tweet deletions (mean= 34.6 tweet deletions per user, S.D.= 56.0). Qualitatively similar results are obtained with less aggressive filtering, e.g., after filtering out the top 2% accounts in total tweet deletions and statuses_count respectively (19M accounts, 1.04B deletions). Since the deletion records span 2011 to 2018, some of the earlier deletion entries and the actual tweets that they reference could have occurred prior to the earliest tweet of an individual user that was used to construct the 26M user network dataset. For example, there could be a deletion record of user A's tweet from 2011, which precedes A's tweets that were collected in 2013 to 2014. As this study reports strictly correlational evidence while prioritizing

coverage, I do not exclude the pre-2015 deletion records on the assumption that a user’s deletion frequency and reasons for deletion are relatively stable over time.

1.3. Political Polarity Data

This study uses a measure of political partisanship on a sample of Twitter users, pre-computed by Garimella and colleagues (43). This measure used in fig. 3 estimates a user’s political polarity based on the aggregate ideological scores of the Twitter users that the focal user follows, ranging from $-\infty$ (liberal) to $+\infty$ (conservative) (44). I matched the numeric user IDs of the 17.9M U.S. Twitter users with tweet deletions in the current study with the IDs in the samples with the polarity scores. This procedure yielded 184K U.S.

1.4. Measures

Network diversity: The canonical measure of network diversity is Burt’s network constraint index. It can be decomposed into the size, density, and hierarchy of an individual’s egocentric network (42, 77). Despite its simplicity and reliability for gauging structural holes from which individuals can reap brokerage benefits, the current study requires separate measurement constructs that reflect different aspects of network diversity due to the subtle countervailing effects – brokerage benefits vs. relational strain – that it can exert on an individual in social media.¹

I use three basic individual-level network measures that reflect different aspects of the diversity of a user’s social network – (a) unweighted node degree (logged), (b) degree-adjusted Shannon entropy of the interaction frequencies across one’s network neighbors (39, 40), and (c) weighted clustering coefficient (41). Node degree k_i is used to gauge the size of user i ’s actively engaged audience. The degree-normalized Shannon entropy D_i quantifies the extent to which i ’s communication is evenly diversified. Specifically,

$$D_i = H_i / \ln k_i$$

where H_i is the Shannon entropy of i ’s relative edge weights:

$$H_i = - \sum_j^{N_i} p_{ij} \ln p_{ij}$$

¹ Furthermore, Everett and Borgatti (77) show that the magnitudes of the size, density, and hierarchy components in network constraint are not necessarily on the same order – for a node’s degree k , size and density are on the orders of $1/k$ whereas hierarchy can range from $1/k^2$ to constant order. The potentially uneven magnitudes of these three components make it tricky to probe which aspect of “diversity” accounts for the variance in post retractions.

where N_i is the set of i 's network neighbors and p_{ij} is the frequency of bidirected mention tweets between i and neighbor j (w_{ij}), relative to the total frequency of mention tweets that i exchanges with all N_i neighbors.

$$p_{ij} = \frac{w_{ij}}{\sum_k^{N_i} w_{ik}}$$

While k_i and D_i pertain to the dyadic communications over which i has substantial control, the weighted clustering coefficient captures the wider community structure in which i is embedded and arguably with less control. It is the portion of communication that node i exchanges with the subset of neighbors who also communicate with one another. Specifically, the weighted clustering coefficient c_i^w is defined as follows:

$$c_i^w = \frac{1}{s_i(k_i - 1)} \sum_{j,h} \frac{(w_{ij} + w_{ih})}{2} a_{ij} a_{ih} a_{jh}$$

where $a_{ij} = 1$ if nodes i and j have an edge (0 otherwise) and s_i is the sum of i 's bidirected edge weights (i.e., weighted degree)

$$s_i = \sum_{j=1}^N a_{ij} w_{ij}$$

These three measures of network diversity each show moderate to high correlations with Burt's network constraint index (Table S1).

1.5. Construct Validity.

Four structural aspects are central to measuring network diversity. The core conceptual objective that the measures should capture is context collapse in network terms. A user with a "diverse" network spans (a) multiple communities, (b) each of similar "importance", and those communities are (c) internally dense (cohesion), but externally exhibit (d) clearly delineated in-group/out-group boundaries (i.e., isolated context). When these conditions are met, I argue that a user will experience more relational strain under context collapse. In graph-theoretic terms, (a) corresponds to the number of communities, roughly mapping onto node degree, while (b) naturally maps on to edge heterogeneity measured by degree-adjusted Shannon entropy. For (c), the weighted clustering coefficient can approximate the dense network communities of a user's egonetwork, but it is a composite measure that cannot by itself distinguish densities of one large, shared context from several those of separate ones. The weighted clustering coefficient is also conceptually (inversely) related to (d), but it cannot capture the finer boundaries between communities. An alternative measure for (c) is the k-core, which captures the highest density subset of neighbors, while (d) can be quantified by the number of egonetwork components. A third possibility that combines the requirements of (c) and (d) is the number of "effective"

communities derived through community detection. These communities optimize for internal density and external sparsity, satisfying both conceptual requirements.

In practice, however, the data show that these alternative measures are highly correlated with node degree – $k\text{-core}=0.81$; number of effective communities= 0.71 (fig. S7). The number of egonetwork components (excluding isolates) showed moderate correlation with degree, but its heavy-tailed distribution makes it sub-optimal. Since the Shannon entropy, degree, and weighted clustering were close to orthogonal to one another and because node degree broadly correlates with alternative measures of (c) and (d), I settled with them. In addition, I used the combination of degree and weighted clustering (interaction terms in the regression models) to approximate conditions (c) and (d) in conjunction based on the premise that for a high degree node to also have high levels of clustering, the ties should have dense interconnections within smaller communities.

2. Political Polarity and Tweet Deletion

Fig. S1 plots the coefficients from an OLS model of tweet deletion frequency for the users whose partisanship was identified (Table S3). Even for these politically active Twitter users who are presumably less hesitant to engage in contentious political debates, the network-structural associations with tweet deletion frequency were, again, positive — and larger than in the full 17.9M sample (Table S3 vs. Table S2) — after adjusting for the user's own ideological position (polarity) and other behavioral measures. Although the moderates had higher levels of network diversity than the partisans (fig. S1A), they tended to delete fewer tweets (fig. S1 inset), which is consistent with previous studies reporting lower tweet deletion rates for moderates and progressives compared to conservatives. The fewer deletions may reflect stronger self-monitoring personality traits (78, 79) or the effects of experiencing the consequences of prematurely posting tweets on hot-button issues before fully considering potential relational repercussions (12, 18).

3. Political Tweet Classification

Political tweet classification of the full corpus (billions of tweets) with state-of-the-art methods is currently infeasible (80). Therefore, the political tweet classification was based on a fine-tuned large language model as described below. First, 4000 randomly sampled tweets, then hand-labeled for political content. A tweet was classified as political if it contained (i) political figures, parties, candidates, and institutions, (ii) elections, governance, policy positions, and government officials, (iii) national and international issues (e.g., healthcare, welfare, and foreign affairs), or (iv) social movements and broad cultural and social issues (e.g., civil rights, moral values, and the environment). Each tweet was labeled by two human annotators based on these criteria. To reduce the inherent content ambiguity in a tweet's short text, the annotators were instructed to look up the hashtags, emoji unicode, and other terms in the tweets that they were uncertain about. A third annotator resolved labeling inconsistencies between the annotators or tweets that an annotator labeled as uncertain.

These 4000 tweets were used for fine-tuning the DeBERTa V3 base model for political tweet classification (81). In the fine-tuning process, before tokenization, each tweet was scanned against a curated year-by-year list of U.S. political keywords (2008–2025); if any keyword matched, the tweet was prefixed with [KW] to give the model an explicit topical cue. Input sequences were truncated or padded to 128 tokens. The Hugging Face Trainer was used for training, with AdamW (Beta_1 = 0.9, Beta_2 = 0.999, Epsilon = 1e-8) and weight decay = 0.01, a linear learning-rate decay schedule (no warm-up), and a per-device batch size of 8. Each epoch was evaluated by weighted F1 to select the best checkpoint. Eventually the model was fine-tuned at a learning rate of 2e-5, achieving 96.65% accuracy (F1=0.9663).

From the polarity-scored users, a random sample of 30,000 accounts was drawn before the activity filters (SI section 1.2); the 25,225 accounts remaining after the filters and their 36 million non-retweet tweets (created 2011–2014) were used for political-tweet classification. Of these tweets, 12.4% were classified political, and 23,905 of the 25,225 accounts had at least one tweet labeled political. Deletions in the data were rare – only 0.018% of the 36 million tweets were deleted (6,582 in total), although this likely underrepresents the true rate by 90%, since the Decahose data from which deletions were tallied are a random 10% of real-time tweets. At the tweet level, political tweets were deleted slightly less often than non-political tweets (0.0122% vs. 0.0192%; 545 vs. 6,037 deleted tweets). At the individual level, 1,722 of the 23,905 accounts deleted at least one tweet in the corpus; averaged across these accounts, the mean per-account probability of deleting a political versus a non-political tweet was 0.36% (S.D. 2.49%) and 0.47% (S.D. 1.50%) respectively, and political tweets made up 7.4% (S.D. 23.5%) of the tweets they deleted. Among the 128 accounts that deleted at least one political *and* one non-political tweet, the within-account geometric-mean odds ratio of political to non-political deletion was 2.24 (95% CI [1.78, 2.84]). For these accounts the mean deletion probabilities were 4.0% (S.D. 7.7%) for political and 1.8% (S.D. 3.6%) for non-political tweets — an absolute difference of 2.2 percentage points — and political tweets rose to 34% (S.D. 20%) of the tweets they deleted. This within-account elevation in political-tweet deletion rests on a thin base — only 545 political deletions, and just 213 accounts with any political deletion (of which 128 also deleted a non-political tweet). Because membership in this subgroup conditions on having deleted at least one tweet of each class, part of this elevated odds ratio is mechanical: under content-blind deletion within each account (i.e., each account's deleted set redrawn at random, with its number of deletions held fixed and contain at least one political and at least one non-political tweet), the expected geometric-mean odds ratio is 2.14 (95% range [1.94, 2.34]; 1,000 permutation replicates), and the observed 2.24 does not exceed this selection null (one-sided permutation $p = 0.17$). The odds ratio therefore describes the subgroup rather than evidencing within-account political targeting on its own.

4. Sentence-BERT Tweet Embedding

4.1. Measures

To measure user-level content characteristics from a user's tweets, I embedded each user's English-language tweets (max 150 tweets) with the all-MiniLM-L6-v2 sentence-transformer (384 dimensions) for 14K randomly sampled users and their egocentric network neighbors from

the 17.9M sample and the 25K users and their egocentric network neighbors drawn from the 184K polarity-score-matched sample.

User Content Atypicality: Atypicality is a measure of a user's position in the embedding space, computed as the cosine distance between the user's mean tweet vector (normalized to unit length) and the centroid of all users' mean vectors. This content atypicality measure highly correlated with the median ego-neighbor embedding distance ($r = 0.85$).

Neighbor-Neighbor Content Diversity: The diversity of content characteristics among a user's neighbors in the sample were quantified by the median pairwise cosine distance of their mean vectors.

Content characteristics: To qualitatively explore a user's actual content characteristic, I summarized the focal users' embeddings by the principal components of their centered mean vectors. The leading components are graded, two-sided axes that, on manual inspection of the tweets at each extreme and corroborated by a separate BERTopic topic model, separate users by the following leading components.

- PC1: interaction purpose (-) personal ↔ (+) networking
- PC2: intended audience (-) intimate ↔ (+) broadcast
- PC3: dynamism (-) affirming-warmth ↔ (+) sports-bravado
- PC4: introspection (-) inspirational ↔ (+) event-hype
- PC5: interactiveness (-) conversational ↔ (+) self-promotional

These first five components captured 41% of variance and provides a qualitative sense of the content characteristics that associate with network diversity and post retraction (Fig. S8 and Model 5 of Table S4).

4.2. Clustering – NN-Embedding Distance Collinearity

Weighted clustering and the median NN-embedding distance are bound to be correlated since both rest on the same information regarding the neighbor-neighbor connections ($r = -0.39$; Table S8). Here, the embedding distance additionally encodes the content information on top of the structural information. Therefore, when these two constructs enter the same regression, the embedding-based measure, which aggregates up to 150 tweets from each of a user's neighbors, will prevail over the weighted clustering coefficient. The attenuation of the clustering terms in Tables S4 and S5 when the content measures enter is therefore the expected outcome under any mechanism operating through this shared audience feature; it is not evidence that network structure is epiphenomenal. Consistent with this view, the degree-adjusted Shannon entropy, which encodes the focal user's interaction information rather than the information between the connected neighbors, is uncorrelated with the content measures ($r \leq 0.05$; Table S8) and retains a positive and significant coefficient in every model of both samples.

4.3. Political Axis Construction

For the political-content measures in the main text, each tweet of the 25K polarity-score-matched users that was submitted for political tweet classification (SI section 3) was embedded separately using all-MiniLM-L6-v2. From this Sentence-BERT embedding, the political axis is the difference between the mean embedding of the political-labeled tweets ($n = 6.5\text{M}$) and the mean embedding of the same users' non-political tweets ($n = 41.6\text{M}$), normalized to unit length. The user-matched contrast prevents the axis from encoding differences between user populations rather than between content classes. A user's political-axis projection is the average of their tweet projections. The per-tweet projection separates political from non-political labels with $\text{AUC} = 0.973$, and the user-level projection correlates 0.937 (Pearson; 0.891 Spearman) with the political tweet share.

5. Agent-Based Model of Opinion Polarization

5.1. Framework

This section details the agent-based model for exploring the conditions under which opinion retraction at the individual level generates opinion polarization at the macro level. The model extends the class of social network opinion dynamics models that have provided the basic mechanistic framework for studying the co-evolution of network segregation and opinion polarization (1, 4, 5, 53, 54). In these models, agents who are similar are more likely to strengthen their relationships to one another (homophily), which, in turn, strengthens their mutual influence on each other's opinion change (social influence). These dyadic processes of homophily and social influence are aggregated across all connected dyads, forming a feedback loop that can generate either polarization or convergence at the macro level, depending on initial conditions and modeling assumptions (52).

These models typically make the following two simplifying assumptions. First, with some exceptions (82), the standard opinion dynamics models assume that the agents' true opinions are fully observable by the alters. This assumption neglects the fact that individuals carefully consider what to express to which audience and self-correct (e.g., retract) to save face or to conform to expectations. Second, models assume that agents are indifferent to the opinion discrepancies between their own network neighbors – typically, third-party relationships do not directly affect the focal agent's own opinion. As a result, fundamental triadic orientations in social interaction are not explicitly incorporated in these models, despite their significance for understanding social behavior (34, 83–85).

Hence, the current model relaxes these simplifying assumptions – an agent decides to retract their posts based on the opinion discrepancies between the alters (alter-alter) in addition to the opinion discrepancies between the focal agent and the alters (ego-alter). This distinction between ego-alter vs. alter-alter relations is important as they may hold different implications for ameliorating opinion retraction and promoting freer speech online – the former points to solutions for fostering interpersonal tolerance and understanding while the latter emphasizes solutions that facilitate flexible audience separation in an open communication environment.

5.2. Model Description

The model simulates information sharing among N agents in a communication network, G with E weighted edges. Each agent i ($i \in N$) has bidirected communication ties to N_i neighbors with tie weight, $w_{ij}^{(t)} \in [0,1]^2$ and holds a time-varying opinion $x_i^{(t)} \in [-1, 1]$ at time t . The agent decides between retracting their post ($Y_i^{(t)} = 0$) that expresses $x_i^{(t)}$ or retain it for their neighbors to see ($Y_i^{(t)} = 1$). At each simulation time step, the agents update these three variables.

5.2.1. Post Retraction Probability

Each agent i posts their opinions at each time step, then decides either to retain their opinion to the network neighbors ($Y_i^{(t)} = 1$) or retract ($Y_i^{(t)} = 0$) with probability $P(Y_i^{(t)})$. The probability of retaining the post at t increases with smaller opinion differences between i and i 's neighbors (ego-alter difference) and among i 's neighbors (alter-alter difference). Specifically,

$$P(Y_i^{(t)} = 1) = 1 - [\beta_a P(X_{i,a}^{(t)}) + (1 - \beta_a) P(X_{i,e}^{(t)})] \quad \text{Eq. 1}$$

$X_{i,a}^{(t)} \in \{0,1\}$ and $X_{i,e}^{(t)} \in \{0,1\}$ indicate conflict occurrence between one or more pairs of i 's alters ($X_{i,a}^{(t)} = 1$) and between i and one or more of i 's neighbors ($X_{i,e}^{(t)} = 1$) at t . $\beta_a \in [0,1]$ is a global parameter which determines the relative importance that the agents place on $P(X_{i,a}^{(t)})$, relative to $P(X_{i,e}^{(t)})$. $P(X_{i,a}^{(t)})$ and $P(X_{i,e}^{(t)})$ are both conflict probabilities compounded by the dyadic opinion differences between their respective agent pairs under consideration. Specifically, the compounded alter-alter conflict probability that agent i considers at time t , given i 's subset of network neighbors $N_i'^{(t-1)} \subset N_i$ whose opinions were observable at $t-1$, is defined by

$$P(X_{i,a}^{(t)}) = 1 - \prod_{j \neq k \in N_i'^{(t-1)}} (1 - q_{i,jk}^{(t)}) \quad \text{Eq. 2}$$

where the product runs over all $n_i^{(t)} = |N_i'^{(t-1)}| \cdot (|N_i'^{(t-1)}| - 1)$ ordered pairs of distinct neighbors whose opinions are observable. $q_{i,jk}^{(t)}$ is the probability, as perceived by observer i , that its neighbors j and k are in conflict at time t . This witnessed-conflict probability is a logistic function of the pair's weighted opinion difference, evaluated against i 's tolerance threshold, $\mu_i^{(t)}$:

$$q_{i,jk}^{(t)} = \frac{1}{1 + e^{-(\alpha_{jk}^{(t)} - \mu_i^{(t)})/\sigma}} \quad \text{Eq. 3}$$

² While N_i does not change after initialization, the tie weights with the neighbors, which are initially assigned from an arbitrary distribution, change through interaction. In addition, while the ties are bidirected the tie weights from i to j need not be equal from j to i .

$$\mu_i^{(t)} = \mu + \sigma \ln n_i^{(t)}, \quad n_i^{(t)} = |N_i^{(t-1)}| \cdot (|N_i^{(t-1)}| - 1) \quad \text{Eq. 4}$$

Here μ is the population baseline tolerance for the opinion difference of a single pair, σ determines the sensitivity of conflict to deviations around the threshold, and $\mu_i^{(t)}$ is the observer's effective tolerance, which rises logarithmically with the number of pair relations the observer monitors. The size adjustment in Eq. 4 is a base-rate (habituation) correction: an agent who interacts with hundreds of neighbors will be exposed to tens of thousands of neighbor pairs and will not treat each dyadic friction as equally alarming. It equalizes the expected number of perceived conflicts in a fully agreeing audience ($\approx e^{-\mu/\sigma}$) across audience sizes, so that the compounded probability in Eq. 2 measures how divided the audience is rather than how large it is; without the adjustment, the product in Eq. 2 approaches one mechanically as the audience grows, regardless of the audience's actual disagreement.

$\alpha_{jk}^{(t)}$ denotes the observed opinion difference between j and k , attenuated by their tie weight $w_{jk}^{(t)}$. Specifically, for neighbor pairs whose opinions were observed in the previous time step,

$$\alpha_{jk}^{(t)} = [\beta_s w_{jk}^{(t)} + (1 - \beta_s)(1 - w_{jk}^{(t)})] \cdot |x_j^{(t)} - x_k^{(t)}| \quad \text{Eq. 5}$$

where $\beta_s \in \{0,1\}$ is a parameter that magnifies the effect of opinion difference on conflict probability between strong-tie neighbors. For example, when $\beta_s = 1$, opinion difference between strong ties (e.g., disagreement between best friends) is maximally intolerable and consequential for conflict, while opinion difference between weak ties matters less. Conversely, when $\beta_s = 0$, opinion differences between weak ties ($w_{jk}^{(t)} \rightarrow 0$) become maximally consequential for conflict.

The compounded ego–neighbor conflict probability takes the same form over the neighbors whose opinions are visible:

$$P(X_{i,e}^{(t)}) = 1 - \prod_{j \in N_i^{(t-1)}} (1 - p_{ij}^{(t)}) \quad \text{Eq. 6}$$

with the dyadic ego–neighbor conflict probability retaining the fixed population baseline tolerance μ and sensitivity σ :

$$p_{ij}^{(t)} = \frac{1}{1 + e^{-(\alpha_{ij}^{(t)} - \mu)/\sigma}} \quad \text{Eq. 7}$$

where $\alpha_{ij}^{(t)}$ follows Eq. 5 applied to the pair (i, j). The asymmetry between Eq. 3 and Eq. 7 is deliberate. An agent's own conflicts accumulate with audience size – the relational strain of addressing a large, context-collapsed audience directly (SI section 5.4) — so the ego–neighbor conflict does not receive the same size adjustment. Observing third-party relationships, by contrast, grows more intense as network size scales, which is why the conflict of each neighbor pair receives the size-adjusted tolerance $\mu_i^{(t)}$. The two dyadic probabilities share the same weighted opinion distance (Eq. 5) and differ only in their tolerance thresholds (μ versus $\mu_i^{(t)}$).

5.2.2. Opinion Update

The current model assumes that an agent's opinion is formed through both attraction to similar opinions and repulsion away from dissimilar opinions (86). Agents update their opinions as a function of the aggregate ego-alter dyadic opinion differences, weighted by tie strength. Specifically, the aggregate influence that the neighbors of agent i exert at time t is,

$$I_i^{(t)} = \frac{1}{2|N_i^{(t-1)}|} \sum_{j \in N_i^{(t-1)}} (2w_{ij}^{(t)} - 1) (x_j^{(t)} - x_i^{(t)}) \quad \text{Eq. 8}$$

In this form, when $2w_{ij}^{(t)} - 1 > 0$, agent i is attracted to agent j 's opinion, but repels away from j 's opinion if $2w_{ij}^{(t)} - 1 < 0$. The aggregate influence is integrated into the following opinion update function (Eq. 7) consisting of the agent's anchor opinion $x_i^{(0)}$ that is initialized at the beginning of the simulation and the agent's currently held opinion, $x_i^{(t)}$. Specifically, agent i 's subsequent opinion is determined by

$$x_i^{(t+1)} = \eta_1 x_i^{(0)} + (1 - \eta_1) [x_i^{(t)} + \eta_2 I_i^{(t)} s_i^{(t)}] \quad \text{Eq. 9}$$

where $\eta_1 \in [0,1]$ is a stubbornness parameter which controls the agent's predisposition to hold its initial anchor opinion (87, 88)(87, 88). The second term, weighted by $1 - \eta_1$, drives the opinion update through a convex combination of the agent's current opinion, $x_i^{(t)}$, and the aggregate influence of the currently observed opinions of i 's neighbors, $I_i^{(t)}$, multiplied by an influence weight parameter $\eta_2 \in [0,1]$, which controls the strength of their influence. Following Flache and Macy (86), a smoothing term $s_i^{(t)}$,

$$s_i^{(t)} = \begin{cases} 1 - x_i^{(t)}, & I_i^{(t)} \geq 0 \\ 1 + x_i^{(t)}, & I_i^{(t)} < 0 \end{cases} \quad \text{Eq. 10}$$

is multiplied to bound the dynamic portion of the opinion update between -1 and 1 (i.e., $-1 \leq x_i^{(t)} + \eta_2 I_i^{(t)} s_i^{(t)} \leq 1$).³ Note that the agent only considers the subset of neighbors $N_i'^{(t-1)} \subseteq N_i$ who posted their opinions in the previous time step.

5.2.3. Tie Weight Update

The model implements continuous over binary edge weights for two reasons. Theoretically, an important aspect of network diversity includes the concentration of interactions at the individual level and have been argued to be unique to each individual as a “social signature” (40). Empirically, edge weight heterogeneity is a central component in the computation of network constraint (89), the canonical measure of network diversity, and approximately 20% of its variance is attributed to edge weight heterogeneity for weighted networks (90). In short, the weights carry significant information that should not be omitted in a model whose mechanism is based on network diversity.

In the model, agent i updates their time-varying directed edge weight to j as,

$$w_{ij}^{(t+1)} = (1 - \gamma) \cdot w_{ij}^{(t)} + \gamma \cdot D_{ij}^{(t)} \quad \text{Eq.11}$$

where $\gamma \in [0,1]$ is a relational update parameter that controls the rate at which opinion differences affect the tie weight and $D_{ij}^{(t)}$ is a sigmoid function that bounds the opinion difference between -1 and 1. Namely,

$$D_{ij}^{(t)} = \frac{1}{1 + e^{-\lambda(1 - |x_j^{(t)} - x_i^{(t)}|)}} \quad \text{Eq.12}$$

5.2.4. Convergence Criteria

Because agents stochastically decide to retract their posts at every step (Eq. 1), the set of opinions visible to each agent is reshuffled each step. Consequently, on degree-heterogeneous (hub) networks and at low stubbornness (low η_1), the microstate—the vector of all agents' opinions—does not freeze even after the population-level opinion distribution has stabilized; it instead fluctuates indefinitely around a stationary distribution. The model therefore typically reaches a dynamic (statistical) equilibrium rather than a frozen fixed point. I detect convergence with two complementary criteria. First, a microstate-freeze test stops the runs that do relax to a

³ $|s_i^{(t)}| \rightarrow 0$ as $|x_i^{(t)}| \rightarrow 1$.

fixed point, triggered when the 95th percentile across agents of the per-step absolute opinion change stays below 0.005 for 20 consecutive steps. Second, for runs that keep churning, I test each run's post-burn-in series of the macro polarization observable (the Esteban–Ray index; (55)) for stationarity, using the augmented Dickey–Fuller (91), KPSS (92, 93), and Geweke (93) tests, with the burn-in (equilibration) point selected by Chodera's method (94) and the effective sample size reported to ensure the steady-state estimate is adequately sampled. A parameter combination is accepted as having reached a dynamic equilibrium when at least 80% of its runs individually pass these stationarity tests, and the reported steady-state quantities are averaged over each run's post-burn-in tail.

5.3. Virtual Experiments

While this opinion retraction model can be used to conduct a broad range of virtual experiments (e.g., network segregation, tie survival) by tuning the full set of model parameters, it is used here to examine the following questions:

- Do agents with diverse networks retract more frequently?
- Do agents with diverse networks hold moderate opinions?
- Do the individual retraction decisions collectively lead to the appearance of heightened polarization?
- Do these decisions also lead to true opinion polarization?

As described in the main text and shown in fig. 6, network structure and initial opinion distribution are the two main experimental manipulations of the current study. The full simulation experiment involves a factorial design with five simulation runs per combination as shown below (59,400 combinations; 297,000 total runs).⁴

- Number of Agents: [200]
- Simulation Steps: [3000]
- Network Type: [completely-connected, small-world, caveman, scale-free, powerlaw-clustered, random]
- Network Density (mean degree $\langle k \rangle$): [≈ 24 , ≈ 48]⁵
- Initial Opinion Distribution: [normal, bimodal, uniform]⁶
- Retraction Enabled: [No, Yes]

⁴ Initial edge weights are drawn from a uniform random distribution between 0 and 1 for all simulations.

⁵ completely-connected network mean degree is fixed at $\langle k \rangle = N - 1 = 199$. Mean degree values are targets that do not necessarily materialize due to the stochasticity of network generators.

⁶ The normal initialization draws opinions with standard deviation 0.5 or 1.0; the bimodal initialization places its two modes at ± 0.46 , ± 0.50 , or ± 0.54 , each with standard deviation 0.1.

- η_1 (stubbornness, Eq. 7): [0.05, 0.10, 0.15]⁷
- η_2 (influence strength, Eq. 7): [0.5, 0.75, 1.0]
- β_s (strong-tie conflict weight Eq. 4): [0.2, 0.8]
- β_a (neighbor-neighbor conflict weight Eq. 1): [0, 0.2, 0.8, 1.0]
- μ (Eq. 3): [0.75, 1.0, 1.5]
- σ (Eq. 3): [0.2]
- γ (Eq. 9): [0.3]
- λ (Eq. 10): [7, 15]
- Proportion of Zealots (fixed-opinion agents): [0, 0.02]

The results from this experiment as reported in figs. 5 and 6 are organized in terms of two individual-level outcomes and two collective outcomes for opinion polarization, which are explained below:

Network diversity and deletion frequency: The first outcome measure is the Pearson correlation between an agent's degree-unadjusted Shannon entropy and frequency of deletion. This correlation is expected to be positive if network brokers (high network diversity) frequently retract their posts to avoid relational conflict. The raw entropy is used here to incorporate the effect of degree as a component of network diversity.

Network diversity and opinion extremity: The second outcome measure is the Pearson correlation between an agent's degree-unadjusted Shannon entropy and the extremity of their opinion. The brokers are expected to be more moderate since their opinions may be influenced by a broader range of diverse opinions. This expectation is borne out in fig. S10: the within-run entropy–extremity correlation averages -0.54 across deletion-enabled runs (negative in 97% of runs) and is nearly invariant across network types (means -0.53 to -0.56), while node degree and clustering are uncorrelated with opinion extremity.

True opinion polarization: I measure true opinion polarization with the Esteban–Ray polarization index (55), defined as

$$ER = \sum_a \sum_b \pi_a^{1+\alpha} \pi_b |y_a - y_b| \quad \text{Eq. 13}$$

⁷ η_1 is heterogeneous across agents: each agent draws its value from a Normal distribution with mean equal to the listed value and standard deviation 0.1, clipped to [0, 1]; the listed values are therefore population means.

where the opinion range $[-1, 1]$ is partitioned into $B = 21$ equal-width bins; π_a and π_b are the fractions of agents in bins a and b , y_a and y_b are the bin midpoints, and $\alpha = 1$ is the polarization-sensitivity parameter. I report the percentage change in this index from the initial to the end state, $(ER_{final} - ER_{initial}) / ER_{initial}$, computed over all agents irrespective of their posting decisions.

- 5 *Observed opinion polarization:* The observed opinion polarization is measured in the same way as described above, but is computed only over the agents who retained their posts at the final time step of the simulation.

5.4. Comparison of Empirical and Simulation Results

- 10 Network diversity and content deletion relate differently in the empirical model (Fig. 3) and the simulation (Fig. 5), and the difference is informative. Both agree that agents with more neighbors delete more, but they diverge on the other two diversity measures: in the Twitter data, more even engagement across neighbors (higher degree-adjusted Shannon entropy) and denser, more clustered neighborhoods accompany more deletion, whereas in the simulation both
15 associations run negative.

- At the most fundamental level, this discrepancy reflects the model assumptions about network diversity. For the purposes of this paper, the model builds in the cost of network diversity by specifying node degree as the source of potential neighbor-neighbor conflict. The retraction rule accumulates disagreement across an agent's neighbors, so the pressure to delete
20 grows directly with how many neighbors it has. Degree is therefore the one structural feature the strain mechanism acts on directly, and it reproduces the empirical pattern that broader exposure brings more relational strain.

- However, the model deliberately omits the countervailing benefits of bridging diverse contacts — the brokerage advantage emphasized in the network-diversity literature — that in
25 real networks rewards diverse, loosely-knit users for staying engaged and can offset the cost of conflict and strain. As a result, the other two non-static diversity measures evolve to reflect only the cost of diversity: agents continually strengthen ties to neighbors they agree with and weaken ties to those they do not, so the evenness of an agent's ties (entropy) and the density of its neighborhood (weighted clustering) end up reflecting how much it agrees with those around it
30 rather than how diverse its engagement is. An agent that agrees broadly keeps many ties active and has little reason to retract, so in the simulation high entropy marks agreement and predicts fewer deletions.

- Likewise, the negative effect of the clustering coefficient reflects the swift and sticky opinion convergence in highly clustered network neighborhoods. Fig. S9, which exhibits varying
35 levels of random rewiring for the Watts-Strogatz small-world network and the caveman network, illustrates this point visually – deletions and true polarization monotonically increases as higher random rewiring probability, p , fades the clustered community structure towards complete randomness. In addition, the within-run entropy–deletion correlation moved monotonically toward the empirical (positive) sign as rewiring dissolved local contexts: in the caveman family
40 the mean correlation rises from -0.212 at $p = 0.3$ to -0.181 at $p = 0.5$ and -0.159 at $p = 0.7$ (OLS slope on p : $+0.133$, $P = 1.4 \times 10^{-84}$), while the small-world family stays essentially flat (-0.196 ,

$-0.185, -0.191$; slope $+0.013$, $P = 0.07$). The divergence between the simulated and empirical entropy associations is therefore largest where intact local community contexts (i.e., the caves) exist and shrinks precisely in the regime that corresponds to the collapsed platform.

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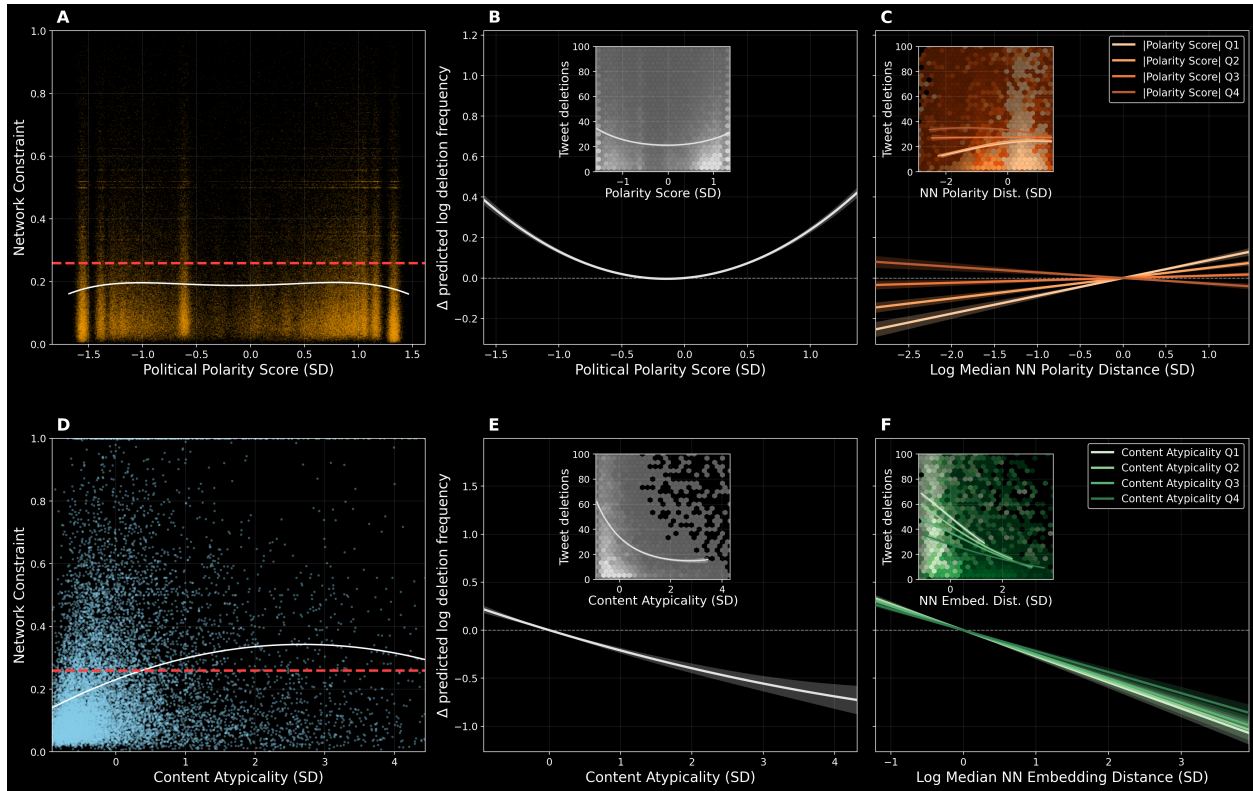


Fig. S1. Network constraint and tweet-deletion marginal effects under two models of a user's own discursive position — political polarity (row 1) and SBERT cognitive atypicality (row 2). **(A–C)** Political-polarity model (Table S3, Model 4; politically-identified users, $N = 79,538$). **(D–F)** SBERT cognitive-atypicality model (Model 4, Table S4; embedding-identified users, $N = 13,879$). **(A, D)** Network constraint versus the focal user's own position: political polarity score (A) and Tweet Atypicality (D) — the ego's cosine distance from the population centroid in SBERT embedding space, in SD units. Points are users; the white curve is a fourth-degree (A) / quadratic (D) polynomial fit; the dashed red line marks the full-sample mean network constraint. **(B, E)** Marginal effect on predicted tweet deletions of polarity score (B; a U-shape — moderates delete least) and Tweet Atypicality (E; declining — more topically atypical users delete less). **(C, F)** Marginal effect of the log median neighbor–neighbor distance — polarity distance (C) and SBERT embedding distance (F) — stratified into |polarity-score| quartiles (C) and content-atypicality quartiles (F). Insets show observed deletions (hexagonal bins, capped at 100) overlaid with a within-stratum smoothed fit; hexagon color encodes the dominant stratum and brightness its local density. Black background; row-2 axes in standardized (SD) units.

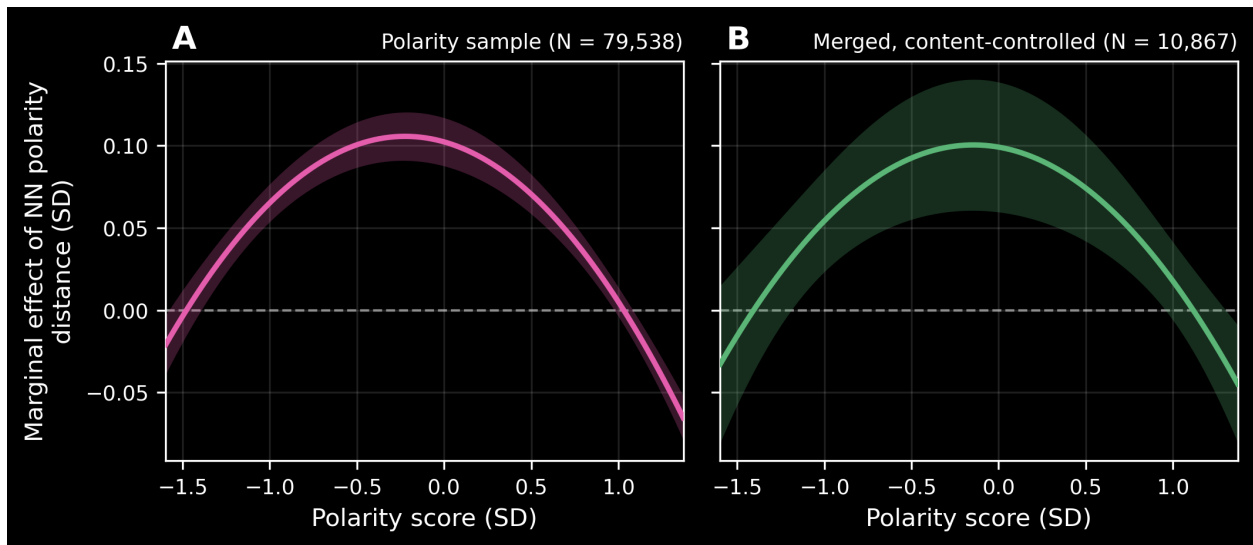


Fig. S2. Marginal effect of the neighbor–neighbor polarity distance on predicted tweet deletions, as a function of the user's polarity score (standardized; 95% CI bands, delta method). (A) Polarity sample (Table S3, Model 4; N = 79,538). (B) The merged polarity–content sample with content atypicality and neighbor content distance controlled (Table S5, Model 4; N = 10,867). The horizontal axis spans the 1st–99th percentile of the standardized polarity score in each estimation sample. In both samples, the partisan dispersion of the audience raises predicted retraction most for politically moderate users and has little effect at the poles.

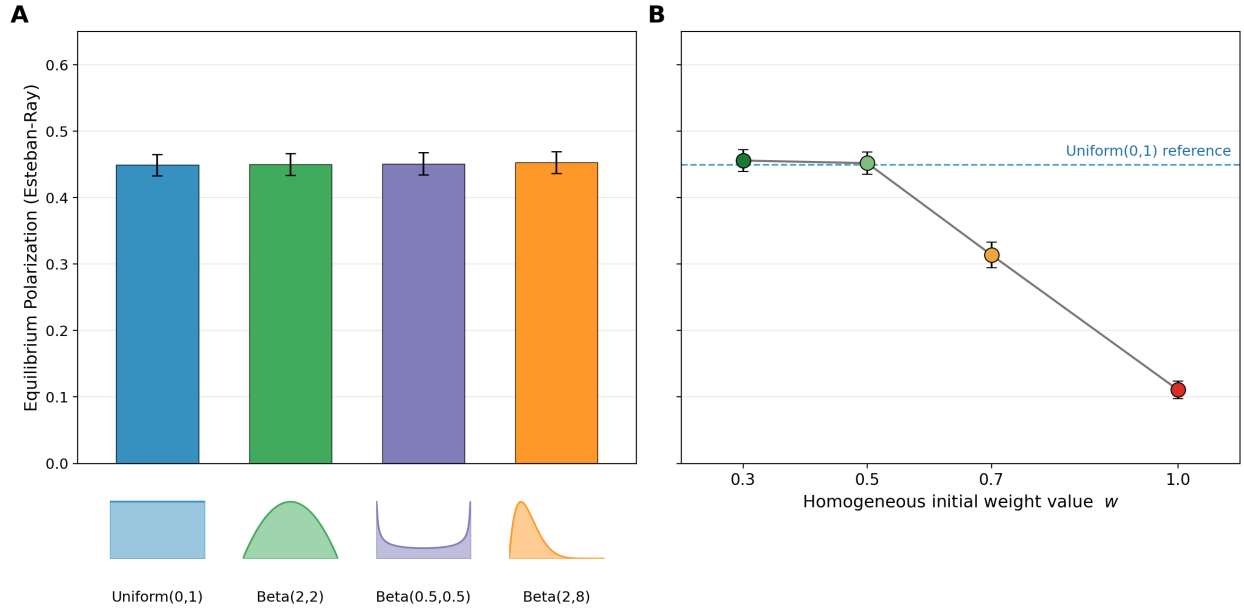


Fig. S3. Polarization is invariant to the initial edge-weight distribution and to homogeneity, provided ties can fall below the neutral value. (A) Equilibrium Esteban–Ray polarization is nearly identical across four heterogeneous initial-weight distributions — Uniform(0,1), Beta(2,2), Beta(0.5,0.5), and Beta(2,8) (insets show each shape; error bars are 95% CIs). (B) With all ties initialized to a single homogeneous value w , polarization matches the uniform reference (dashed) for $w \leq 0.5$ and collapses to consensus only as $w \rightarrow 1$ — the unweighted/binary limit, where the influence kernel $(2w-1)$ becomes pure attraction. Means over six networks and eight parameter cells ($\eta_1 \times \eta_2 \times \mu$), deletion enabled, normal initial opinions.

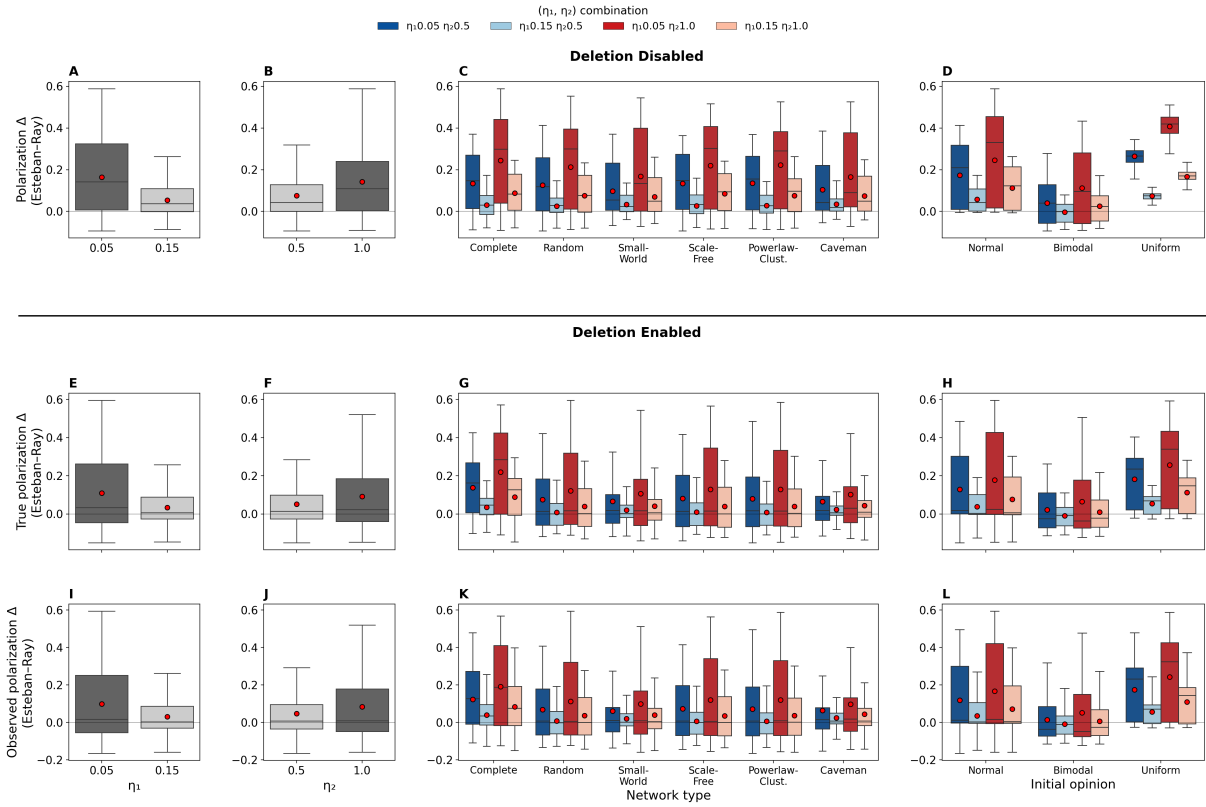


Fig. S4. Sensitivity of polarization to the initial-opinion anchor strength (η_1) and social-influence strength (η_2). Change in Esteban–Ray polarization (final – initial) with deletion disabled (top block; true $\equiv$ observed) and deletion enabled (bottom block; true opinions E–H, observed opinions I–L). Columns show the η_1 main effect, the η_2 main effect, and the four (η_1 , η_2) combinations across network type and initial-opinion distribution. Polarization increases as the $x(0)$ anchor weakens (η_1 : 0.15 $\rightarrow$ 0.05) and influence strengthens (η_2 : 0.5 $\rightarrow$ 1.0), consistently across all networks and initial distributions. Proportion of zealots in the population= 2%.

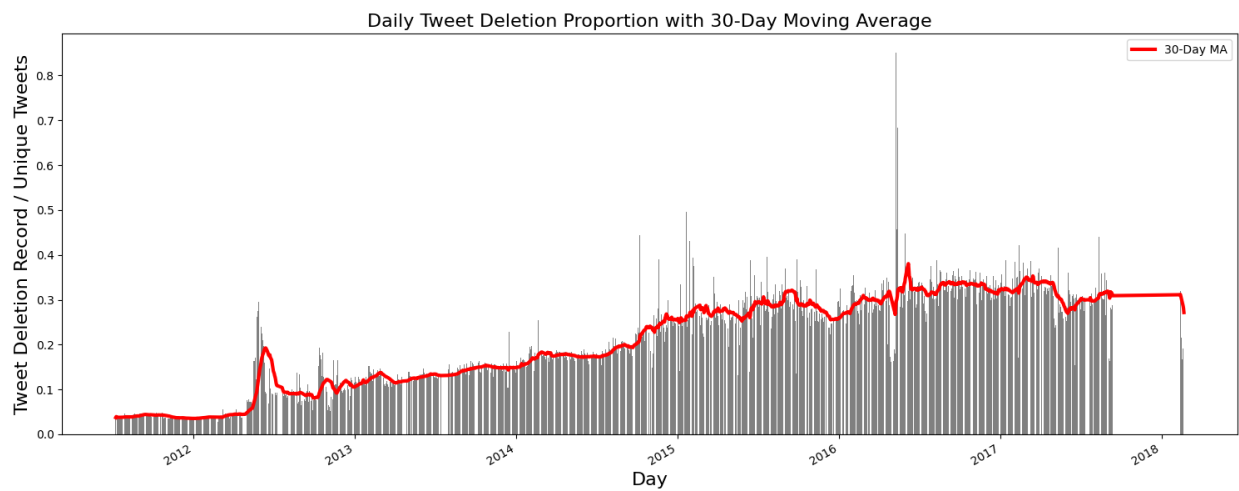


Fig. S5. Proportion of daily tweet deletion records, relative to the total number of unique tweets (i.e., non-retweets) streamed through the Decahose.

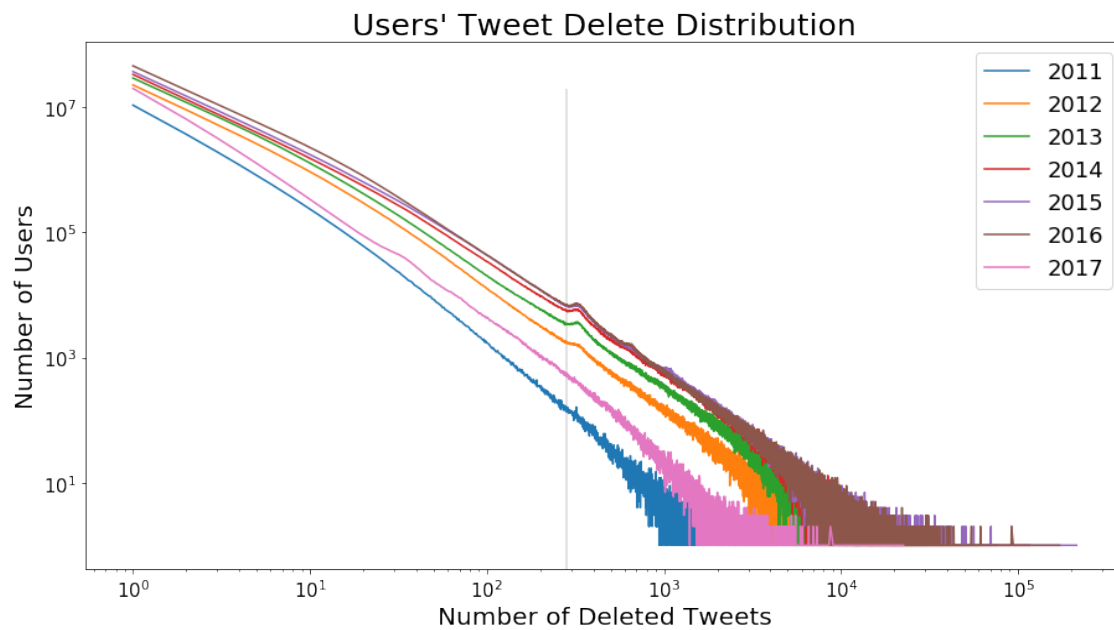


Fig. S6. Yearly user-level tweet deletion distribution.

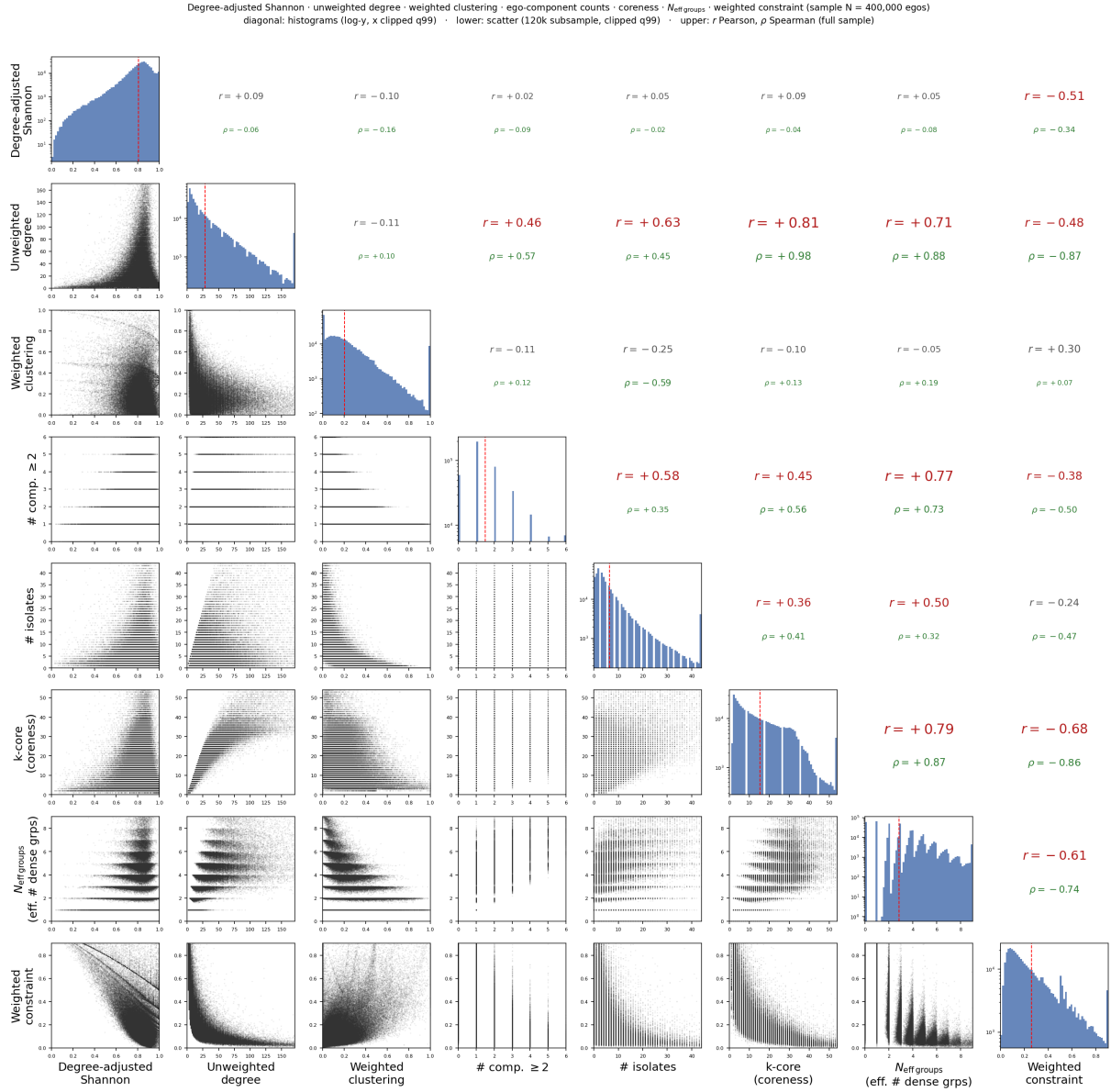


Fig. S7. Scatter-plot matrix of the network-diversity measures and alternatives. Pairwise relationships among degree-adjusted Shannon entropy of tie strengths, unweighted node degree, weighted clustering, number of ego components ≥ 2 , number of isolates, k-core coreness, effective number of dense neighbor communities, and Burt's weighted network constraint (random sample $N = 400,000$). Diagonal: histograms (log-frequency y-axis; x-axis clipped at the 99th percentile). Lower triangle: pairwise scatter (120,000-user subsample; axes clipped at the 99th percentile). Upper triangle: Pearson r (top) and Spearman ρ (bottom), computed on the full sample.

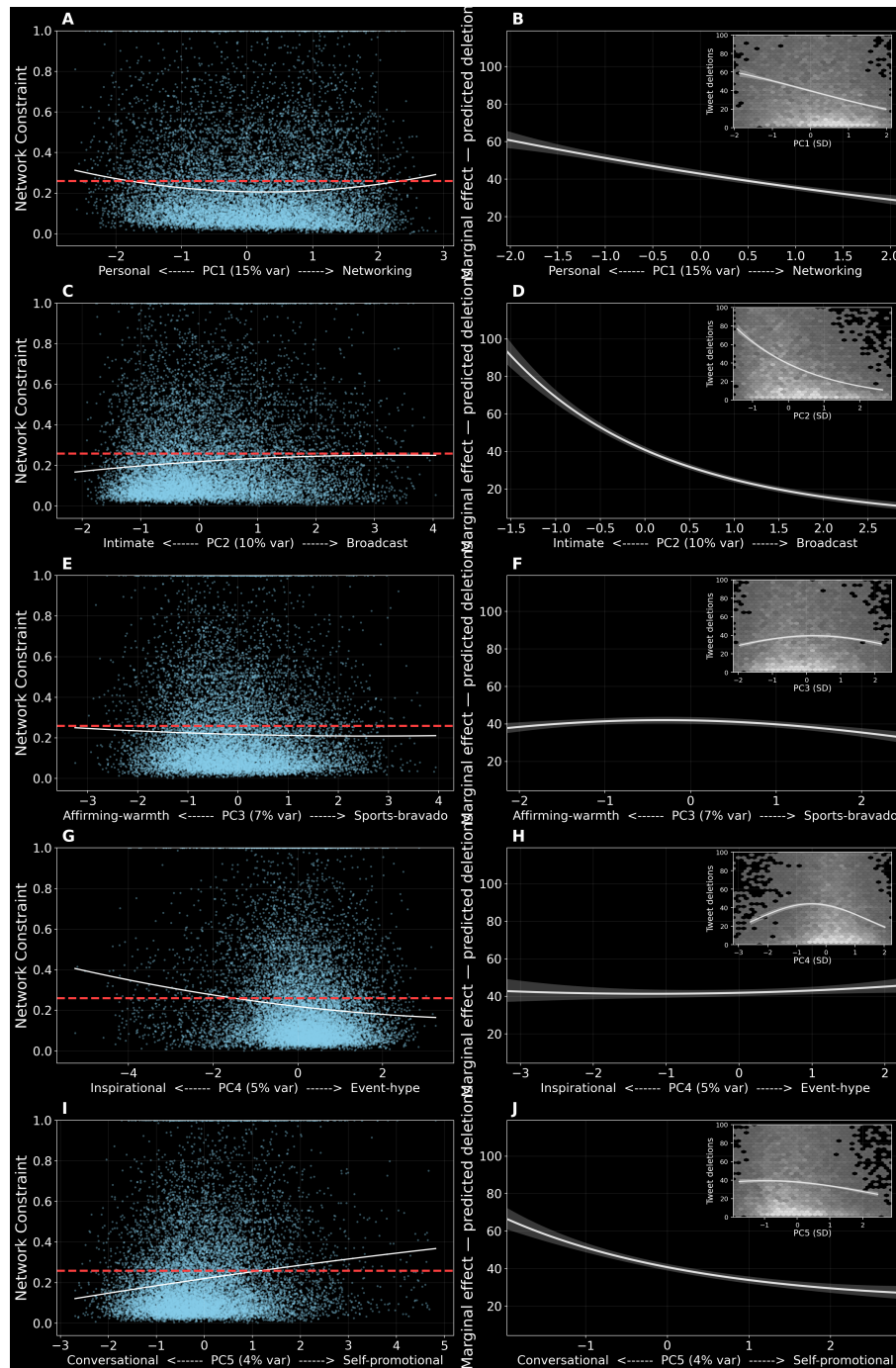


Fig. S8. The focal user's position on each of the five leading SBERT embedding components (PC1–PC5). Column A: network constraint versus the PC position (points are users; white quadratic fit; dashed red = mean constraint of the full 17.9M sample). Column B: marginal effect of the PC position on predicted tweet deletions (95% band; within-stratum smoothed inset). Each axis runs from its A pole (negative) to its B pole (positive): PC1 Personal→Networking (15% var), PC2 Intimate→Broadcast (10%), PC3 Affirming-warmth→Sports-bravado (7%), PC4 Inspirational→Event-hype (5%), PC5 Conversational→Self-promotional (4%). Personal/intimate/conversational poles delete more; networking/broadcast/promotional poles delete less. X axes are in standard deviation units.

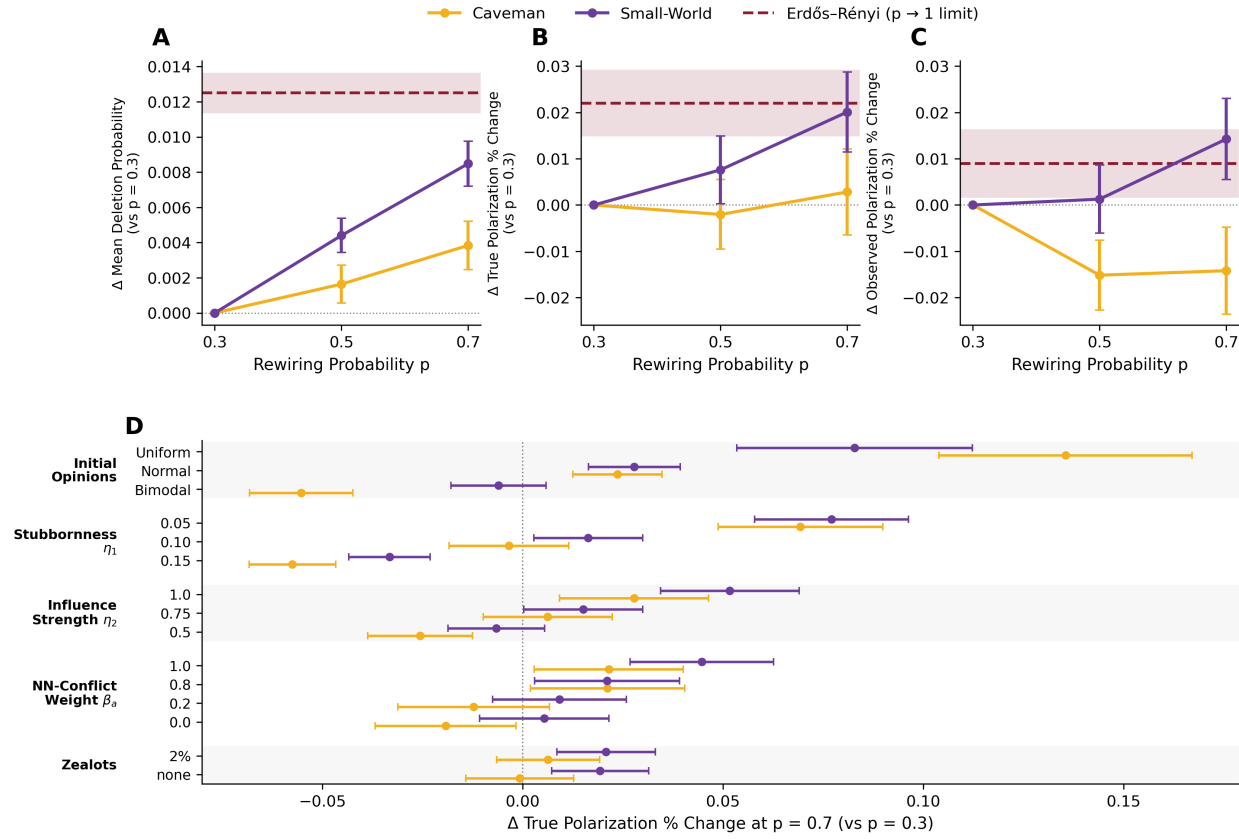


Fig. S9. Dissolving local contexts by rewiring raises retraction and true polarization at fixed audience size, converging toward the degree-matched random-graph limit. The factorial sweep is identical at every rewiring level, so each of the 5,184 parameter combinations per network family (deletion-enabled runs; five runs per combination averaged) is observed at rewiring probability $p = 0.3$, $p = 0.5$ and $p = 0.7$. (A–C) Mean within-combination change relative to the same family's $p = 0.3$ baseline (95% CI across combinations) for mean deletion probability (A) and true (B) and observed (C) Esteban–Ray polarization % change, for the caveman (yellow; 8 caves of 25 agents) and Watts–Strogatz small-world (purple; $k = 24$) topologies. The burgundy dashed line and band give the Erdős–Rényi limit ($p=1$). (D) The same within-combination change in true polarization at $p = 0.7$, by strata of the main factorial axes: initial opinion distribution, stubbornness η_1 (Eq. 7), influence strength η_2 (Eq. 7), neighbor–neighbor conflict weight β_a (Eq. 1), and zealot share.

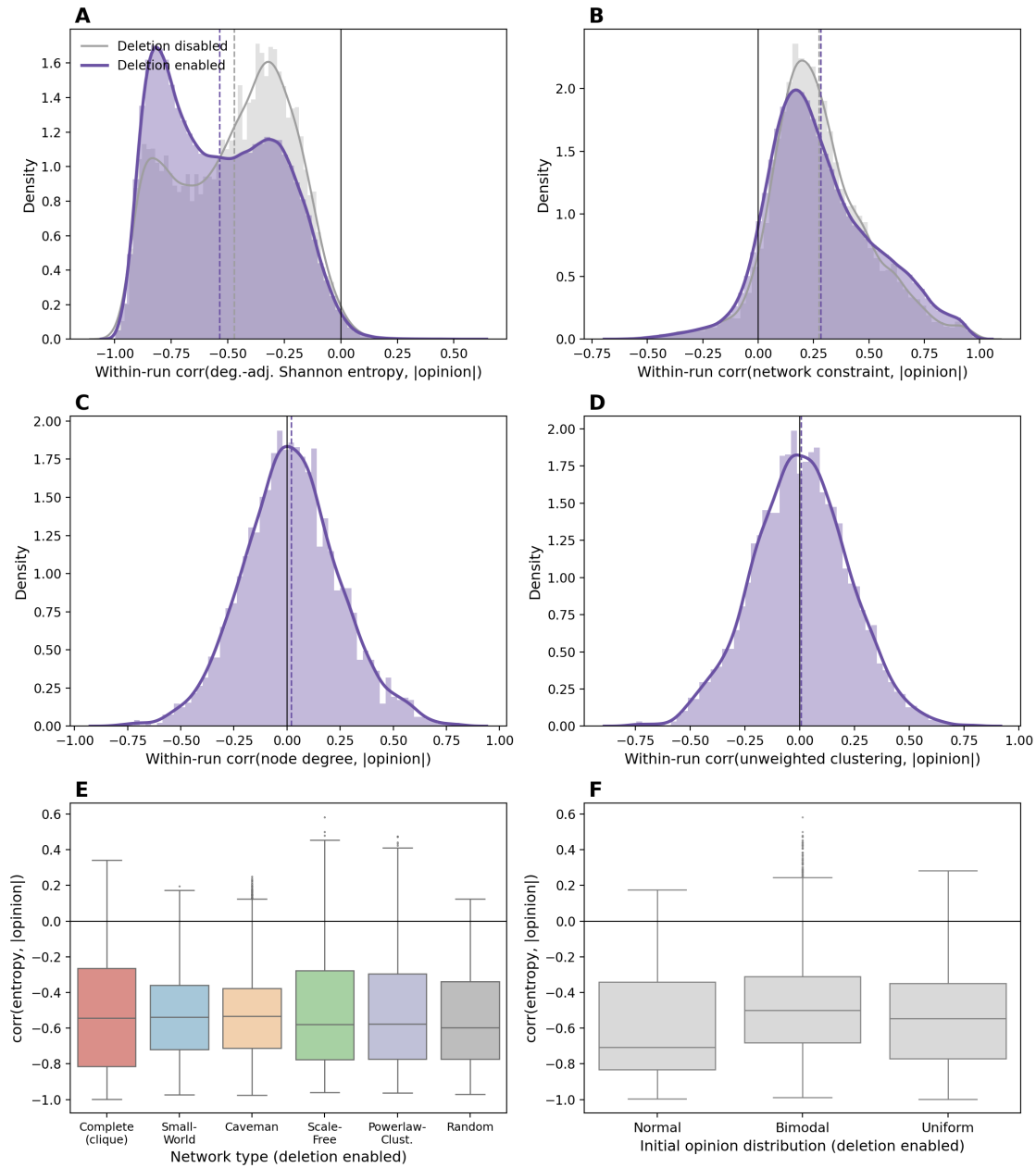


Fig. S10. Network diversity and opinion moderation in the agent-based model. Within-run Pearson correlations between an agent's network position and its final-step opinion extremity $|x|$. (A) Degree-adjusted Shannon entropy of tie weights: deletion-enabled runs (purple; $N = 285,120$ runs; mean $r = -0.54$, negative in 97% of runs) vs. deletion-disabled runs (gray; $N = 11,880$; mean -0.47); dashed lines mark means. (B) Burt's network constraint, (C, D) Node degree and unweighted clustering coefficient, computed on the pooled deletion-enabled agent sample of Table S7. (E, F) The entropy–extremity correlation by network type and by initial opinion distribution (deletion-enabled runs). Agents with diverse (high-entropy, low-constraint) engagement hold systematically more moderate opinions in every topology, most strongly under normal and least under bimodal initialization, whereas audience size and local clustering are unrelated to opinion extremity.

Table S1. Pearson correlation of network diversity measures.

	Degree-Adjusted Shannon Entropy	Log Degree	Weighted Clustering	Weighted network constraint
Degree-Adjusted Shannon Entropy				
Log Degree	0.06			
Weighted Clustering	-0.10	-0.13		
Weighted network constraint	-0.50	-0.77	0.30	

5 **Table S2.** Log tweet deletion frequency OLS model All control and network-diversity terms are standardized (z-scored); coefficients are in SD units and interactions are de-correlated from their main effects (main effects read at the sample mean).

	Model 1		Model 2	
	B	t	B	t
Intercept	2.647	9753.14	2.662	9719.57
Log Status Count	0.709	1706.78	0.714	1723.72
Log Followers	0.146	324.41	0.155	345.21
Log Friends	-0.085	-218.12	-0.079	-203.30
Degree-Adjusted Shannon Entropy	0.030	106.33	0.037	128.23
Log Degree	-0.001	-2.14	0.004	10.45
Weighted Clustering	0.101	353.34	0.196	491.73
Log Degree × Weighted Clustering			0.109	340.79
N	17,895,941		17,895,941	
Adjusted R²	0.292		0.297	

Table S3. Log tweet deletion frequency OLS models for Twitter users whose polarity scores were measured. The control, network-diversity, and polarity terms are standardized (z-scored) — the network and control terms on the 17.9M-user sample, the polarity terms on the polarity-identified sample; coefficients are in SD units and interactions are de-correlated from their main effects (main effects read at the sample mean).

	Model 1		Model 2		Model 3		Model 4	
	B	t	B	t	B	t	B	t
Intercept	2.717	834.42	2.581	542.46	2.304	264.55	2.299	263.86
Log Status Count	0.794	189.9	0.785	188.15	0.902	136.99	0.898	136.55
Log Followers	0.022	5.9	0.023	6.21	-0.082	-18.09	-0.083	-18.38
Log Friends	-0.086	-24.69	-0.086	-24.62	-0.031	-6.64	-0.036	-7.65
Deg.-Adj. Shannon Entropy	0.068	24.09	0.065	22.99	0.088	18.49	0.087	18.2
Log Degree	0.1	27.37	0.106	29.06	0.15	24.33	0.153	24.7
Weighted Clustering	0.156	39.19	0.166	41.61	0.072	9.13	0.073	9.31
Log Deg x Weighted Clust	0.071	21.21	0.078	23.29	0.068	10.12	0.07	10.38
Polarity Score			0.013	5.08	0.066	18.2	0.06	16.45
Polarity Score Squared			0.142	39.08	0.2	38.33	0.198	37.82
Log Median Neighbor-Neighbor Polarity Dist					0.014	3.76	0.102	13.67
Polarity x NN Polarity Dist							-0.03	-8.85
Polarity ² x NN Polarity Dist							-0.067	-12.91
N	184,753		184,753		79,538		79,538	
Adjusted R2	0.2941		0.3001		0.3001		0.3019	

Table S4. Standardized log tweet-deletion frequency OLS — the SBERT cognitive (main exploratory) model. Outcome $\ln(\text{deletions}+1)$ kept raw, so B = log-deletions per 1 SD. Each model reports B and t.

	Model 1		Model 2		Model 3		Model 4		Model 5	
	B	t	B	t	B	t	B	t	B	t
Intercept	3.237	283.05	3.278	290.21	3.315	293.98	3.302	281.66	3.353	289.79
Log Status Count	0.676	44.53	0.627	41.91	0.609	41.26	0.612	41.43	0.518	33.02
Log Followers	0.332	19.44	0.386	22.97	0.372	22.43	0.368	22.18	0.324	19.77
Log Friends	-0.158	-10.75	-0.144	-10.03	-0.114	-8.03	-0.113	-7.94	-0.094	-6.81
Degree-Adjusted Shannon Entropy	0.04	3.46	0.045	4.05	0.027	2.47	0.025	2.21	0.043	3.94
Log Degree	-0.056	-3.9	-0.138	-9.56	-0.196	-13.47	-0.203	-13.88	-0.076	-4.74
Weighted Clustering	0.182	12.75	0.118	8.34	0.014	0.97	0.007	0.46	0.001	0.1
Log Degree x Weighted Clustering	0.12	9.49	0.075	6.0	0.043	3.44	0.037	2.96	0.017	1.44
Content Atypicality			-0.251	-25.65	-0.16	-15.05	-0.184	-15.13	0.216	8.17
Content Atypicality Squared			0.072	7.39	0.043	4.44	0.024	2.3	-0.023	-2.19
Log Median NN Embedding Distance					-0.228	-20.51	-0.249	-20.31	-0.137	-10.82
Atypicality x NN Embedding Distance							0.032	4.07	0.026	3.42
PC1 Position (Personal-Networking)									-0.17	-14.81
PC2 Position (Intimate-Broadcast)									-0.471	-23.89
PC3 Position (Affirming-Bravado)									-0.021	-2.27
PC4 Position (Inspirational-Hype)									-0.002	-0.13
PC5 Position (Conversational-Promotional)									-0.188	-11.76
N	13,879		13,879		13,879		13,879		13,879	
Adjusted R2	0.2605		0.2956		0.3163		0.3171		0.364	

Table S5. Standardized OLS estimates of log tweet-deletion frequency on network diversity, content characteristics, and partisan audience dispersion. Sample: the 30K polarity-score-matched U.S. users (partisan positions inferred from the political accounts they follow) after the activity filters (deletion count ≤ 350 ; statuses $\leq 20,000$): N = 25,204 (Model 1) and 25,202 (Model 2); Models 3–4 are restricted to the N = 10,867 users with at least two polarity-scored neighbors, as required by the neighbor–neighbor polarity distance. Content measures are constructed from Sentence-BERT (all-MiniLM-L6-v2) mean tweet vectors of the focal users and their neighbors (SI section 4); the ego–neighbor content distance is excluded for collinearity with content atypicality ($r = 0.85$). Interaction terms are products of the standardized components, following Table S3.

	Model 1		Model 2		Model 3		Model 4	
	B	t	B	t	B	t	B	t
Intercept	2.575	200.73	2.532	200.34	2.335	100.01	2.329	99.67
Log Status Count	0.818	72.64	0.799	72.26	0.915	52.13	0.911	51.89
Log Followers	0.071	6.88	0.067	6.61	-0.047	-3.62	-0.048	-3.69
Log Friends	-0.094	-10.0	-0.053	-5.65	-0.002	-0.17	-0.007	-0.52
Deg-Adj. Shannon Entropy	0.074	9.89	0.054	7.3	0.066	5.12	0.064	5.0
Log Degree	0.054	5.09	-0.041	-3.82	0.005	0.31	0.009	0.49
Weighted Clustering	0.168	15.71	0.001	0.04	-0.033	-1.49	-0.031	-1.42
Deg. x Weighted Clust.	0.077	8.63	0.023	2.59	-0.022	-1.25	-0.02	-1.16
Polarity Score	0.017	2.44	0.022	3.23	0.073	7.68	0.069	7.16
Polarity Score Squared	0.139	14.26	0.145	15.16	0.185	13.4	0.186	13.37
Content Atypicality	-0.107	-14.79	-0.059	-8.18	-0.056	-5.42	-0.056	-5.38
Content Atypicality Squared	-0.036	-5.46	-0.052	-8.05	-0.025	-2.59	-0.025	-2.54
Log Median NN Embedding Dist.			-0.262	-32.54	-0.238	-16.28	-0.237	-16.18
Log Median NN Polarity Distance					0.018	1.8	0.099	4.92
Polarity Score x NN Polarity Distance							-0.018	-2.0
Polarity Score ² x NN Polarity Distance							-0.063	-4.53
N	25,204		25,202		10,867		10,867	
Adjusted R2	0.3163		0.3438		0.3294		0.3306	

Table S6. Standardized OLS estimates of log tweet-deletion frequency with political-content moderation. Sample: the merged polarity–content sample of Table S5 (N = 25,202 in all models). All models retain the full covariate set of Table S5 Model 2. The diversity index is the Table S5 Model 1–implied linear combination of the four standardized network terms ($0.054 \cdot \log \text{degree} + 0.168 \cdot \text{weighted clustering} + 0.077 \cdot \text{degree} \times \text{clustering} + 0.074 \cdot \text{Shannon entropy}$), re-standardized on the estimation sample; because the index is exactly linear in terms already present, only its interactions enter the models. Political tweet share = fraction of the user’s classified tweets labeled political (SI section 3); political-axis projection = mean per-tweet projection on the political axis (SI section 4.3).

	Model 1		Model 2		Model 3		Model 4	
	B	t	B	t	B	t	B	t
Intercept	2.502	192.7	2.502	192.59	2.498	191.47	2.486	191.39
Log Status Count	0.796	72.14	0.796	72.14	0.795	72.05	0.797	72.45
Log Followers	0.075	7.44	0.075	7.42	0.076	7.48	0.076	7.5
Log Friends	-0.05	-5.32	-0.05	-5.32	-0.049	-5.28	-0.048	-5.14
Degree-Adjusted Shannon Entropy	0.056	7.63	0.057	7.62	0.051	6.71	0.06	8.08
Log Degree	-0.058	-5.33	-0.058	-5.34	-0.07	-6.1	-0.065	-6.02
Weighted Clustering	-0.022	-1.84	-0.022	-1.84	-0.039	-2.96	-0.036	-3.04
Log Degree x Weighted Clustering	0.005	0.58	0.005	0.53	-0.016	-1.39	-0.005	-0.5
Polarity Score	0.027	3.93	0.027	3.93	0.027	3.95	0.031	4.48
Polarity Score Squared	0.158	16.12	0.158	16.12	0.159	16.19	0.165	16.9
Content Atypicality	-0.046	-6.14	-0.046	-6.13	-0.046	-6.14	-0.031	-4.11
Content Atypicality Squared	-0.059	-9.01	-0.059	-9.02	-0.06	-9.12	-0.062	-9.45
Log Median NN Embedding Distance	-0.252	-30.98	-0.252	-30.92	-0.251	-30.87	-0.239	-29.1
Political Tweet Share	-0.063	-8.27	-0.063	-8.21	-0.064	-8.35		
Political Tweet Share x Diversity Index	-0.061	-8.02	-0.061	-7.5	-0.056	-6.83		
Political-Axis Projection							-0.101	-12.65
Political-Axis Projection x Diversity Index							-0.077	-9.62
Content Atypicality x Diversity Index			-0.002	-0.34	-0.008	-1.02	0.01	1.29
Diversity Index x Log Status Count					0.025	2.92		
Diversity Index x Polarity Score					-0.028	-4.22		
N	25,202		25,202		25,202		25,202	
Adjusted R ²	0.3466		0.3466		0.3472		0.3495	

Table S7. Simulation results. agent deletion-probability regression. Within-run fixed effects; cluster-robust t by run. 97,260 agents across 4800 runs. Coefficients are in within-run standard-deviation units. Model 1 uses weighted clustering. Model 2 uses unweighted (topological) clustering, fixed at network initialization.

	Model 1		Model 2	
	B	t	B	t
Degree-Adjusted Shannon Entropy	-0.162	-30.38	-0.164	-30.83
Degree	0.481	91.26	0.472	90.02
Weighted Clustering	-0.016	-4.56		
Degree $\times$ Weighted Clustering	-0.021	-6.12		
Unweighted Clustering			-0.040	-12.27
Degree $\times$ Unweighted Clustering			-0.027	-7.86
Opinion Score x (signed)	-0.031	-7.52	-0.031	-7.47
Opinion Score Squared	0.099	22.59	0.098	22.55
Median Neighbor-Neighbor Opinion Distance	0.098	22.07	0.096	21.45
Opinion Score $x \times$ Neighbor-Neighbor Opinion Distance	0.008	2.38	0.009	2.49
Opinion Score Squared $\times$ Neighbor-Neighbor Opinion Distance	-0.017	-4.72	-0.017	-4.71
N	97,260		97,260	
Within-run R^2	0.327		0.328	

Table S8. Pearson correlations among network diversity, polarity, and SBERT content measures on the merged polarity+SBERT sample (N=25,202; all off-diagonal $p < 0.001$ except Log Degree–Polarity Score, $r = -0.004$, $p = 0.57$). Sample = intersection of the 30K polarity focal set (deletions ≤ 350 and statuses $\leq 20,000$ caps) with users having SBERT embeddings. Log Degree = $\ln(\text{unweighted degree})$; Log Median NN Embedding Distance = $\log(\text{median} + 1)$; Atypicality = standardized ego-centroid cosine distance over focal SBERT means; Polarity = Barberá score.

	Degree-Adjusted Shannon Entropy	Log Degree	Weighted Clustering	Polarity Score	Atypicality	Log Median NN Embedding Distance
Degree-Adjusted Shannon Entropy	1.000	0.067	-0.189	0.028	0.045	0.022
Log Degree	0.067	1.000	-0.107	-0.004	-0.199	-0.281
Weighted Clustering	-0.189	-0.107	1.000	-0.023	-0.062	-0.391
Polarity Score	0.028	-0.004	-0.023	1.000	0.048	0.047
Atypicality	0.045	-0.199	-0.062	0.048	1.000	0.293
Log Median NN Embedding Distance	0.022	-0.281	-0.391	0.047	0.293	1.000